



本教學投影片係屬教科書著作之延伸，亦受著作權法之保護。

請依出版社授權使用規範利用本教學資源，非經授權許可不得以影印、拷貝、列印等方法進行重製，亦不得改作、公開展示著作內容，亦請勿對第三人公開傳輸、散布。



Chapter 5

Synchronous Sequential Logic

Prof. Chih-Cheng Tseng (曾志成)

cctseng@mail.ntou.edu.tw

Chapter Objectives

- Know how to distinguish a sequential circuit from a combinational circuit.
- Understand the functionality of an *SR* latch, transparent latch (or *D* latch), *D* flip-flop, *JK* flip-flop, and *T* flip-flop.
- Know how to use the characteristic table and characteristic equation of a flip-flop.
- Know how to derive the state equation, state table, and state diagram of a clocked sequential circuit.
- Know the difference between Mealy and Moore finite state machines.
- Given the state diagram of a finite state machine, be able to write a HDL model of the machine.

Chapter Objectives

- Understand the HDL models of latches and flip-flops.
- Know how to write synthesizable HDL models of clocked sequential circuits.
- Know how to design a state machine using manual methods.
- Know how to eliminate equivalent states in a state table.
- Know how to define a one-hot state assignment code.
- Be able to design a sequential circuit with (a) *D* flip-flops, (b) *JK* flip-flops, and (c) *T* flip-flops.

5.1 Introduction

- Combinational circuits
 - Contain no memory elements
 - The outputs depend only on the inputs

5.2 Sequential Circuits (1/4)

- Sequential circuits

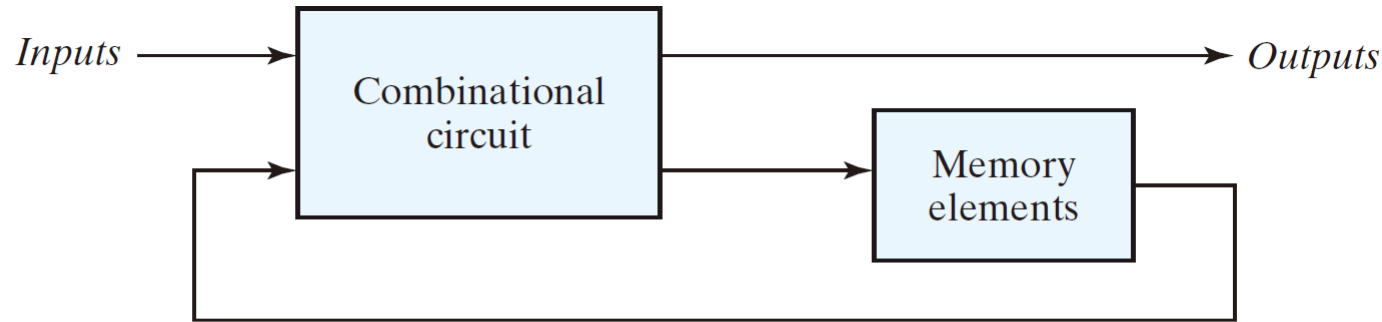


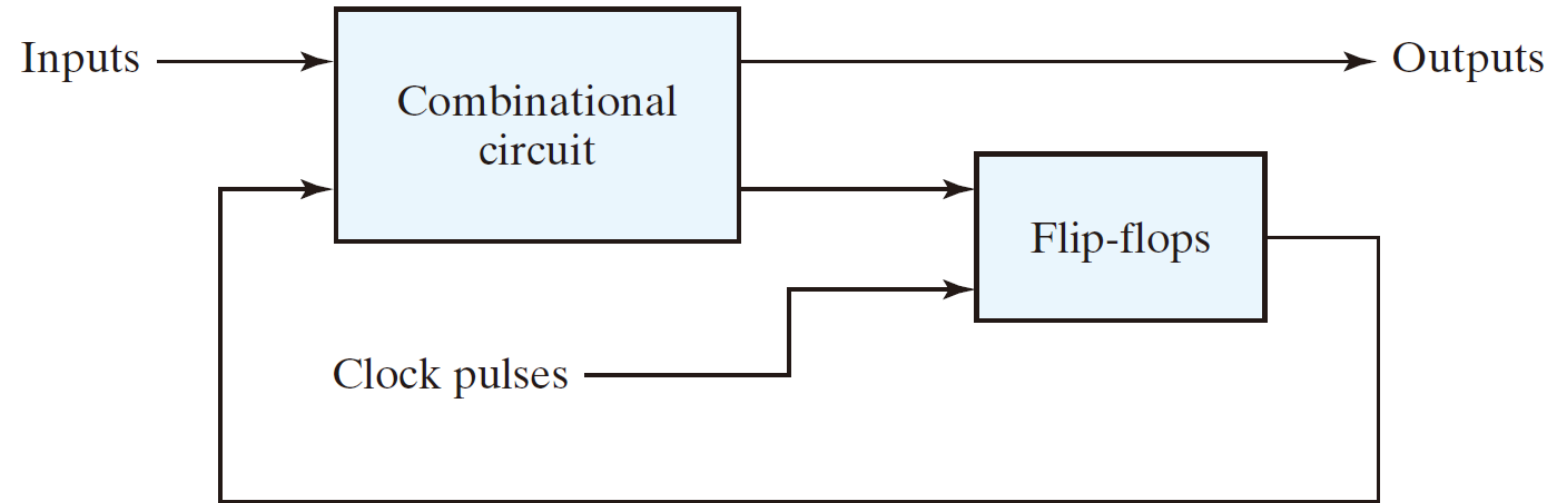
FIGURE 5.1
Block diagram of sequential circuit

- A feedback path
- The state of the sequential circuit
- $(\text{inputs, current state}) \Rightarrow (\text{outputs, next state})$
- Synchronous: the transition happens at discrete instants of time
- Asynchronous: the transition happens at any instant of time

5.2 Sequential Circuits (2/4)

- Synchronous sequential circuits
 - Most commonly used
 - A master-clock generator to generate a periodic train of clock pulses
 - The clock pulses are distributed throughout the system
 - Clocked sequential circuits
 - No instability problems
 - The memory elements: flip-flops (正反器)
 - Binary cells capable of storing one bit of information
 - Two outputs: one for the normal value and one for the complement value
 - Maintain a binary state indefinitely until directed by an input signal to switch states

5.2 Sequential Circuits (3/4)



(a) Block diagram



(b) Timing diagram of clock pulses

FIGURE 5.2
Synchronous clocked sequential circuit

5.2 Sequential Circuits (4/4)

- **Practice Exercise 5.1** Describe the fundamental difference between the output of a combinational circuit and the output of a sequential circuit.

Answer:

1. The output of a combinational circuit depends on only the inputs to the circuit.
2. The output of a sequential circuit depends on the inputs to the circuit and the present state of the storage elements.

5.3 Latches (門鎖器) (1/8)

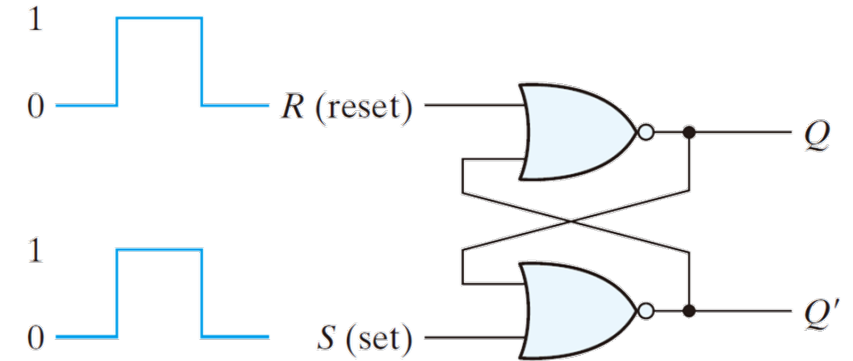
- Latch
 - Storage element that operates with signal levels rather than signal transitions
 - Level-sensitive device
- Flip-flop
 - Constructed by the latches
 - Storage element that is controlled by a clock transition
 - Edge-sensitive device

5.3 Latches (2/8)

SR Latch

- Implementation approach #1: Two cross-coupled NOR gates

- An asynchronous sequential circuit
- Two inputs
 - S for set
 - R for reset
- Two mutually complementary outputs Q and Q'
- Functionalities
 - No operation
 - $(S, R) = (0, 0) \Rightarrow (Q(t+1), Q'(t+1)) = (Q(t), Q'(t))$
 - Reset
 - $(S, R) = (0, 1) \Rightarrow (Q(t+1), Q'(t+1)) = (0, 1)$
 - Set
 - $(S, R) = (1, 0) \Rightarrow (Q(t+1), Q'(t+1)) = (1, 0)$
 - Forbidden
 - $(S, R) = (1, 1) \Rightarrow (Q(t+1), Q'(t+1)) = (0, 0)$



(a) Logic diagram

S	R	Q	Q'
1	0	1	0
0	0	1	0 (after $S = 1, R = 0$)
0	1	0	1
0	0	0	1 (after $S = 0, R = 1$)
1	1	0	0 (forbidden)

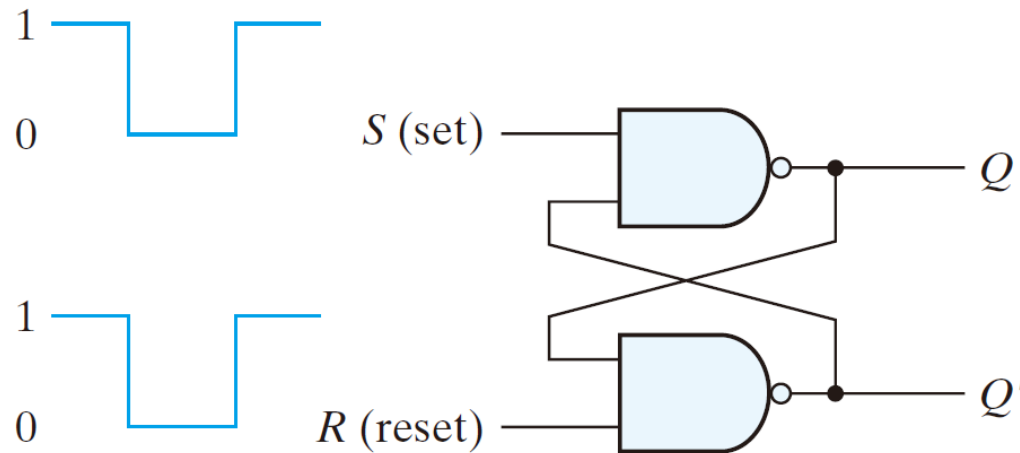
(b) Function table

FIGURE 5.3
SR latch with NOR gates

5.3 Latches (3/8)

SR Latch

- Implementation approach #2: Two cross-coupled NAND gates



(a) Logic diagram

S	R	Q	Q'
1	0	0	1
1	1	0	1 (after $S = 1, R = 0$)
0	1	1	0
1	1	1	0 (after $S = 0, R = 1$)
0	0	1	1 (forbidden)

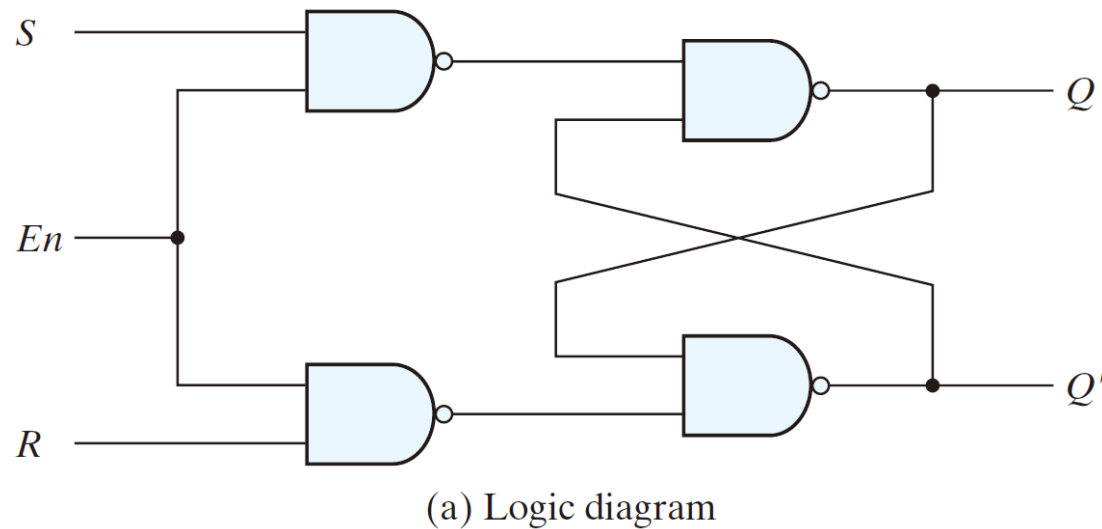
(b) Function table

FIGURE 5.4
SR latch with NAND gates

5.3 Latches (4/8)

SR Latch

- *SR* latch with control input
 - $En = 0$, no change
 - $En = 1$, NOR-gates-based *SR* latch



En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5
SR latch with control input

5.3 Latches (5/8)

SR Latch

- **Practice Exercise 5.2**

(a) What input condition puts an *SR* NOR latch into an indeterminate state?

Answer: Both inputs are 1.

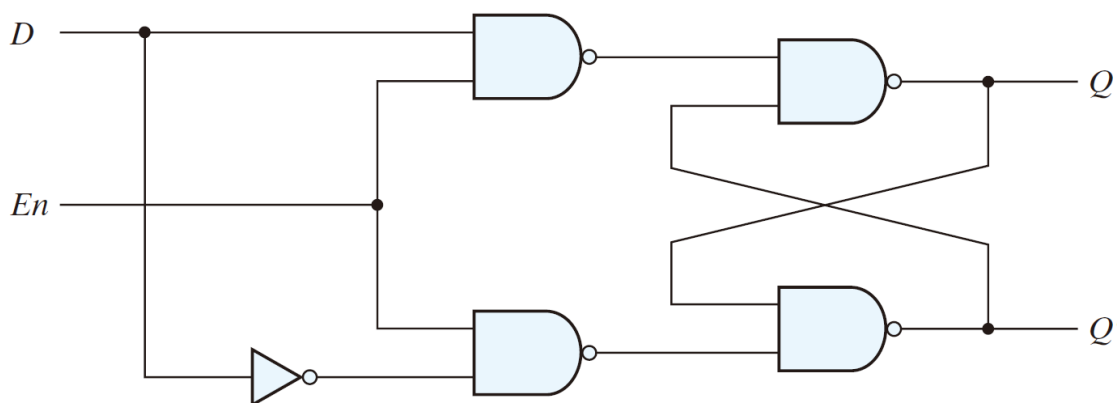
(b) What input condition puts an *SR* NAND latch into an indeterminate state?

Answer: Both inputs are 0.

5.3 Latches (6/8)

D Latch

- *D* Latch (Transparent Latch)
 - Eliminate the undesirable conditions of the indeterminate state in the *SR* latch
 - *D* for data
 - Gated *D*-latch
 - $D \Rightarrow Q$ (or $Q = D$) when $En = 1$; no change (or $Q(t + 1) = Q(t)$) when $En = 0$



(a) Logic diagram

En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

(b) Function table

FIGURE 5.6
D latch

5.3 Latches (7/8)

D Latch

- Graphic Symbols

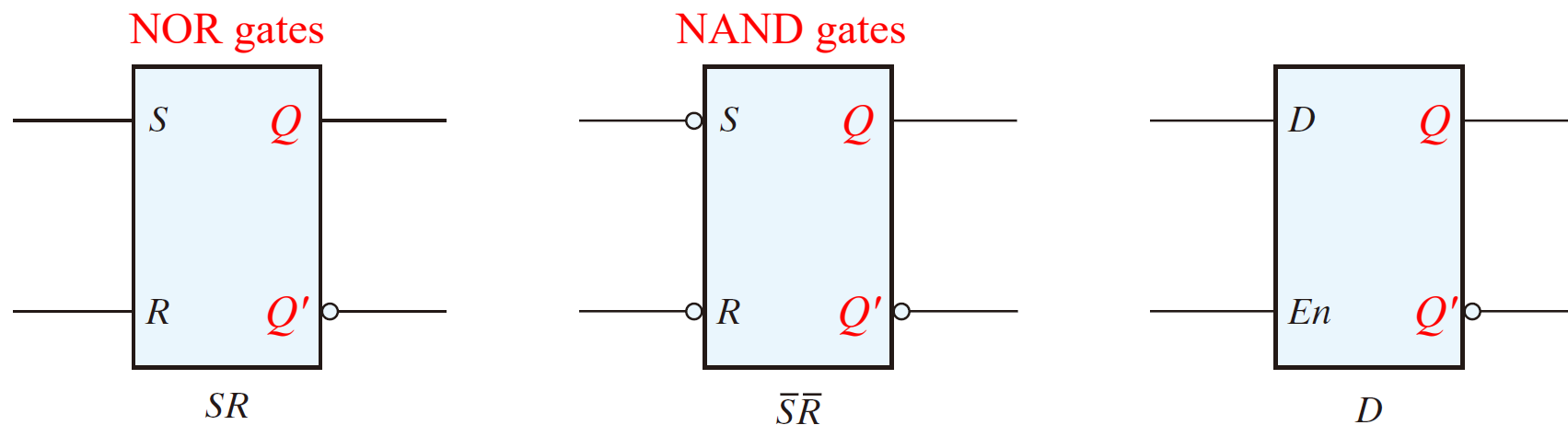


FIGURE 5.7
Graphic symbols for latches

5.3 Latches (8/8)

D Latch



- **Practice Exercise 5.3** Describe the functionality of a transparent latch.

Answer:

1. A transparent latch has a data input, an enable input, and two outputs.
2. When the enable input is asserted, the output Q follows the input to the latch.
3. When the enable input is de-asserted, the output Q is held at the value that was present at the moment the enable input was de-asserted.

5.4 Storage Element: Flip-Flops (1/16)

- A trigger (觸發) is a momentary change of the control input that causes a switch of the state of a latch or flip-flop.
 - Level-triggered – latches
 - Edge-triggered – flip-flops



(a) Response to positive level



(b) Positive-edge response



(c) Negative-edge response

FIGURE 5.8

Clock response in latch and flip-flop

5.4 Storage Element: Flip-Flops (2/16)

- If level-triggered flip-flops are used
 - the feedback path may cause instability problem
- Edge-triggered flip-flops
 - the state transition happens only at the edge
 - eliminate the multiple-transition problem

5.4 Storage Element: Flip-Flops (3/16)

Edge-Triggered D Flip-Flop

- Implementation approach #1: Leader-follower D flip-flop
 - Two separate D latches
 - A leader D latch (positive-level triggered)
 - A follower D latch (negative-level triggered)

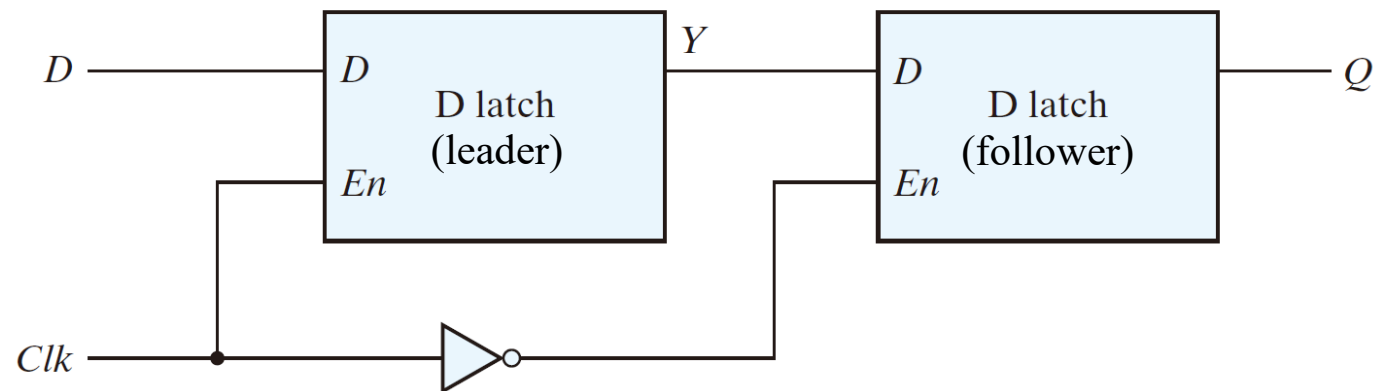


FIGURE 5.9
Leader-follower D flip-flop

5.4 Storage Element: Flip-Flops (4/16)

Edge-Triggered *D* Flip-Flop

- Implementation approach #2: Edge-triggered flip-flops
 - the state changes during a clock-pulse transition
- A *D*-type positive-edge-triggered flip-flop

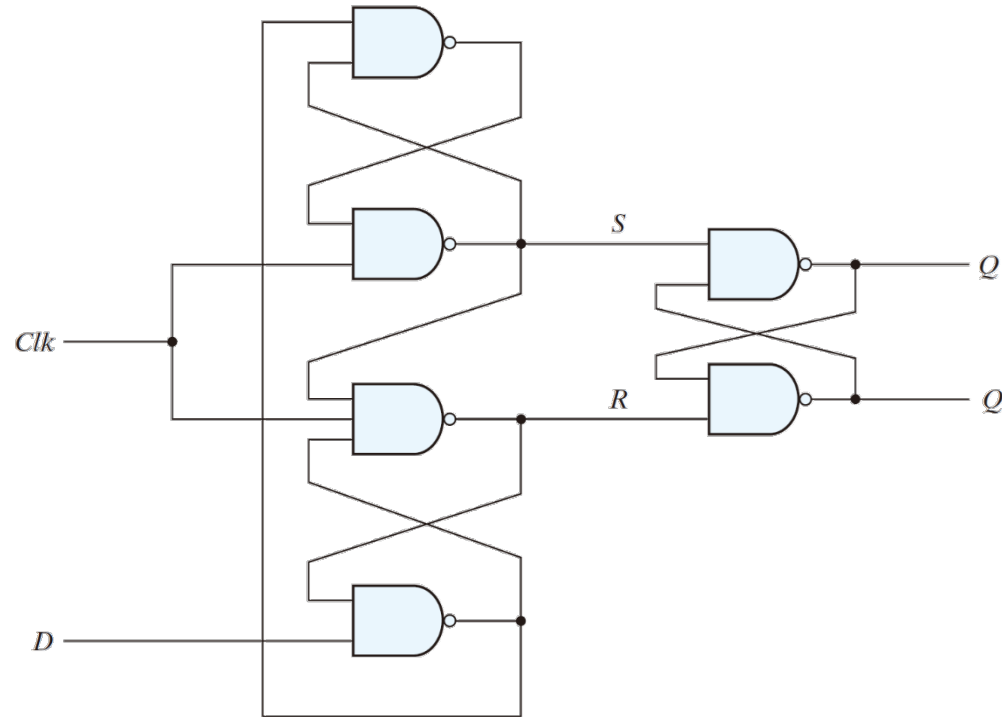


FIGURE 5.10
D-type positive-edge-triggered flip-flop

5.4 Storage Element: Flip-Flops (5/16)

Edge-Triggered *D* Flip-Flop

- Three basic *SR* latches
 - $(S, R) = (0, 1): Q = 1$
 - $(S, R) = (1, 0): Q = 0$
 - $(S, R) = (1, 1):$ no operation
 - $(S, R) = (0, 0):$ should be avoided

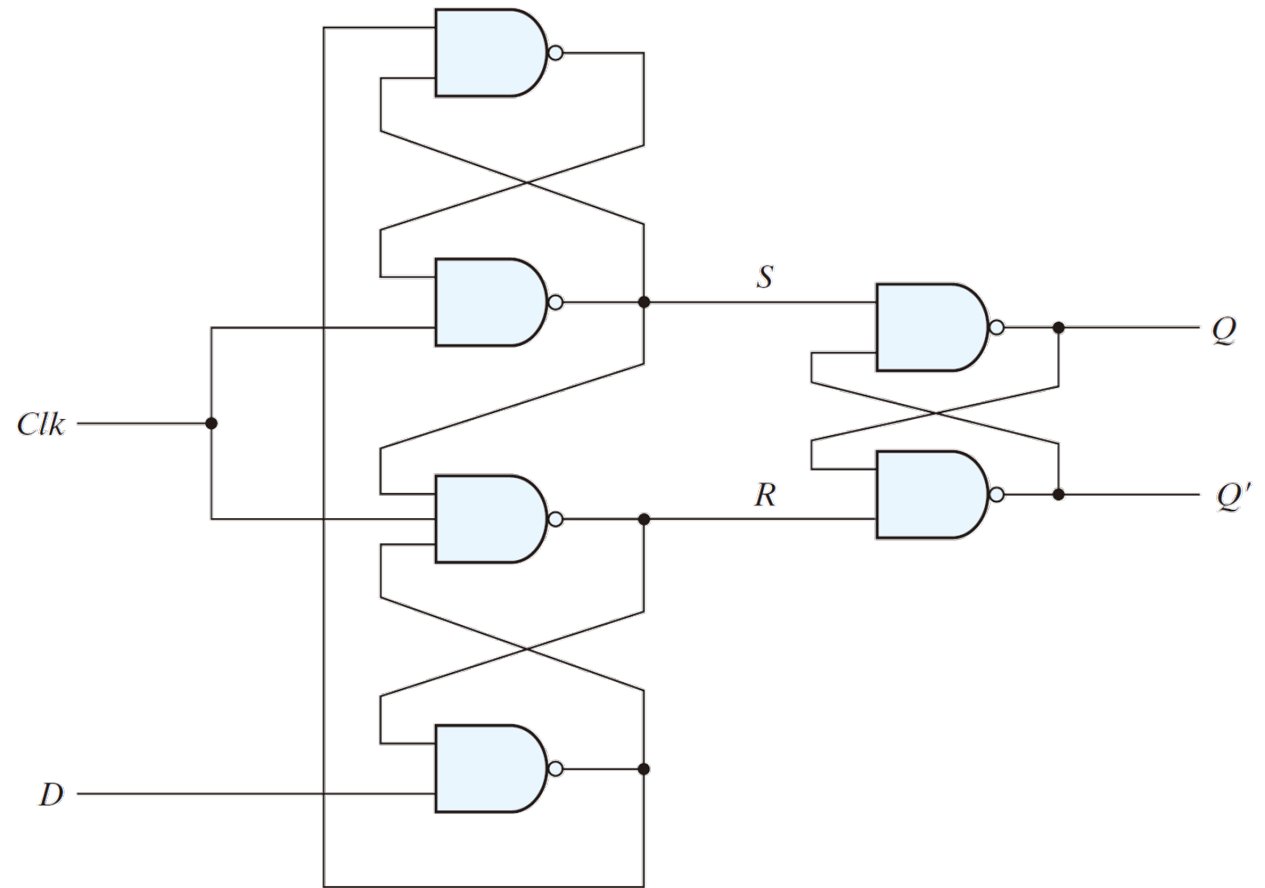


FIGURE 5.10
D-type positive-edge-triggered flip-flop

5.4 Storage Element: Flip-Flops (6/16)

Edge-Triggered D Flip-Flop

- The setup time
 - D input must be maintained at a constant value prior to the occurrence of the positive clock Clk pulse.
 - Data to the internal latches
- The hold time
 - D input must not change after the application of the positive Clk pulse
 - Clock to the internal latch
- The propagation delay time of the flip-flop is defined as the interval between the trigger edge and the stabilization of the output to a new state.

5.4 Storage Element: Flip-Flops (7/16)

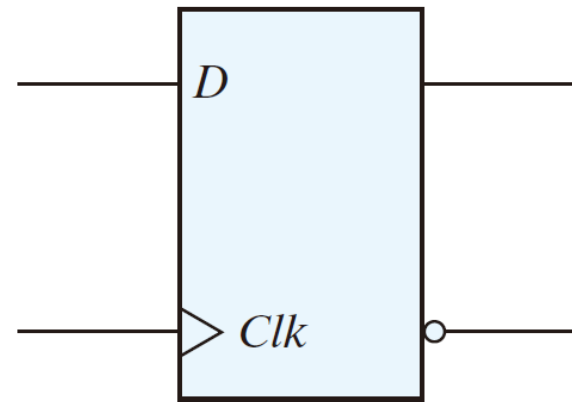
Edge-Triggered D Flip-Flop

- Summary
 - $Clk = 0$: $(S, R) = (1, 1)$, no state change
 - $Clk = \uparrow$: state change once
 - $Clk = 1$: state holds
 - eliminate the feedback problems in sequential circuits
- All flip-flops must make their transition at the same time

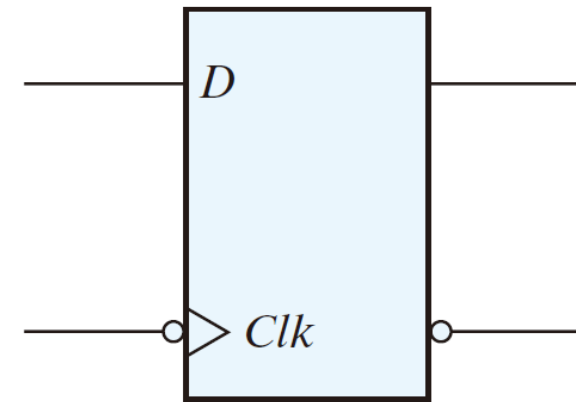
5.4 Storage Element: Flip-Flops (8/16)

Edge-Triggered D Flip-Flop

- Graphic symbols
- The edge-triggered D flip-flops
 - The most economical and efficient
 - Positive-edge and negative-edge



(a) Positive-edge



(a) Negative-edge

FIGURE 5.11

Graphic symbol for edge-triggered D flip-flop

5.4 Storage Element: Flip-Flops (9/16)

Edge-Triggered D Flip-Flop

- **Practice Exercise 5.4** What is meant by “a positive-edge flip-flop?”
Answer: A positive-edge flip-flop is one that is activated by the rising (positive) edge of the clock (synchronizing signal).

5.4 Storage Element: Flip-Flops (10/16)

Other Flip-Flops --- *JK* Flip-Flop

- $D = JQ'(t) + K'Q(t)$
 - $J = 0, K = 0: D = Q(t) \Rightarrow Q(t + 1) = Q(t)$ (no change)
 - $J = 0, K = 1: D = 0 \Rightarrow Q(t + 1) = 0$
 - $J = 1, K = 0: D = 1 \Rightarrow Q(t + 1) = 1$
 - $J = 1, K = 1: D = Q'(t) \Rightarrow Q(t + 1) = Q'(t)$

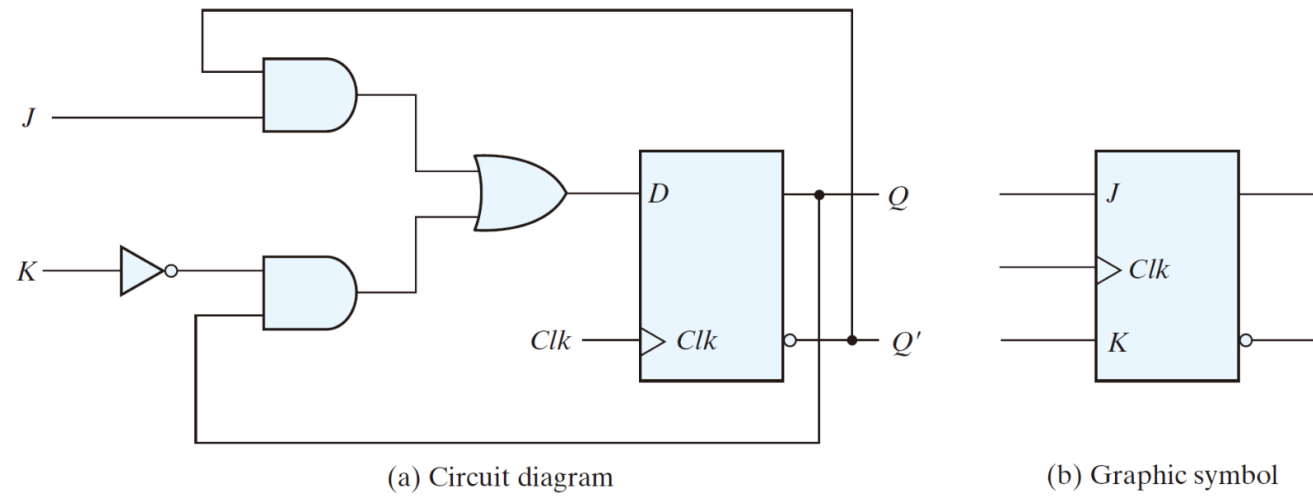
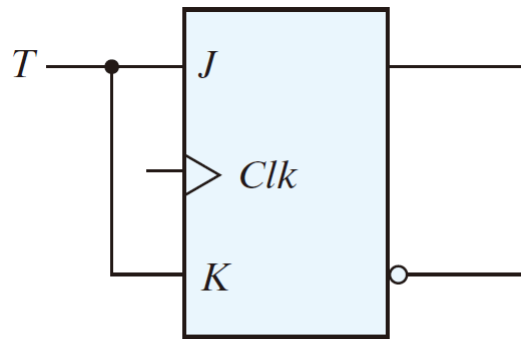


FIGURE 5.12
JK flip-flop

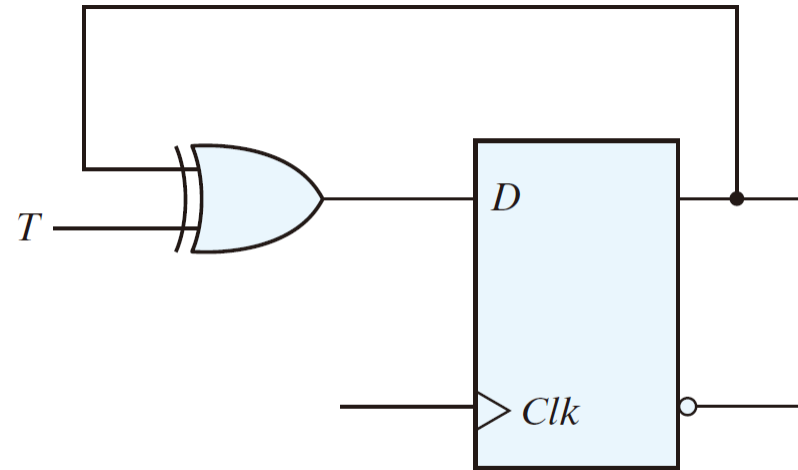
5.4 Storage Element: Flip-Flops (11/16)

Other Flip-Flops --- T Flip-Flop

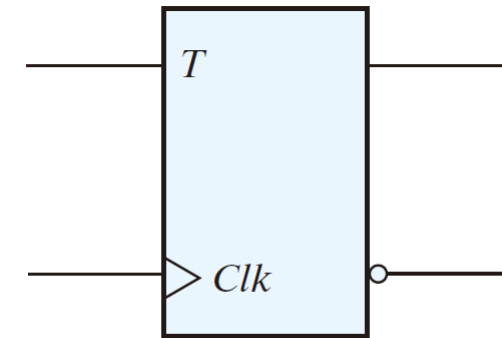
- $D = T \oplus Q(t) = TQ'(t) + T'Q(t)$
 - $T = 0: D = Q(t) \Rightarrow Q(t+1) = Q(t)$ (no change)
 - $T = 1: D = Q'(t) \Rightarrow Q(t+1) = Q'(t)$



(a) From JK flip-flop



(b) From D flip-flop



(c) Graphic symbol

FIGURE 5.13
 T flip-flop

5.4 Storage Element: Flip-Flops (12/16)

Other Flip-Flops

- Characteristic Tables (特徵表，特性表)

Table 5.1
Flip-Flop Characteristic Tables

<i>JK Flip-Flop</i>			
<i>J</i>	<i>K</i>	<i>Q(t + 1)</i>	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

<i>D Flip-Flop</i>		
<i>D</i>	<i>Q(t + 1)</i>	
0	0	Reset
1	1	Set

<i>T Flip-Flop</i>		
<i>T</i>	<i>Q(t + 1)</i>	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

5.4 Storage Element: Flip-Flops (13/16)

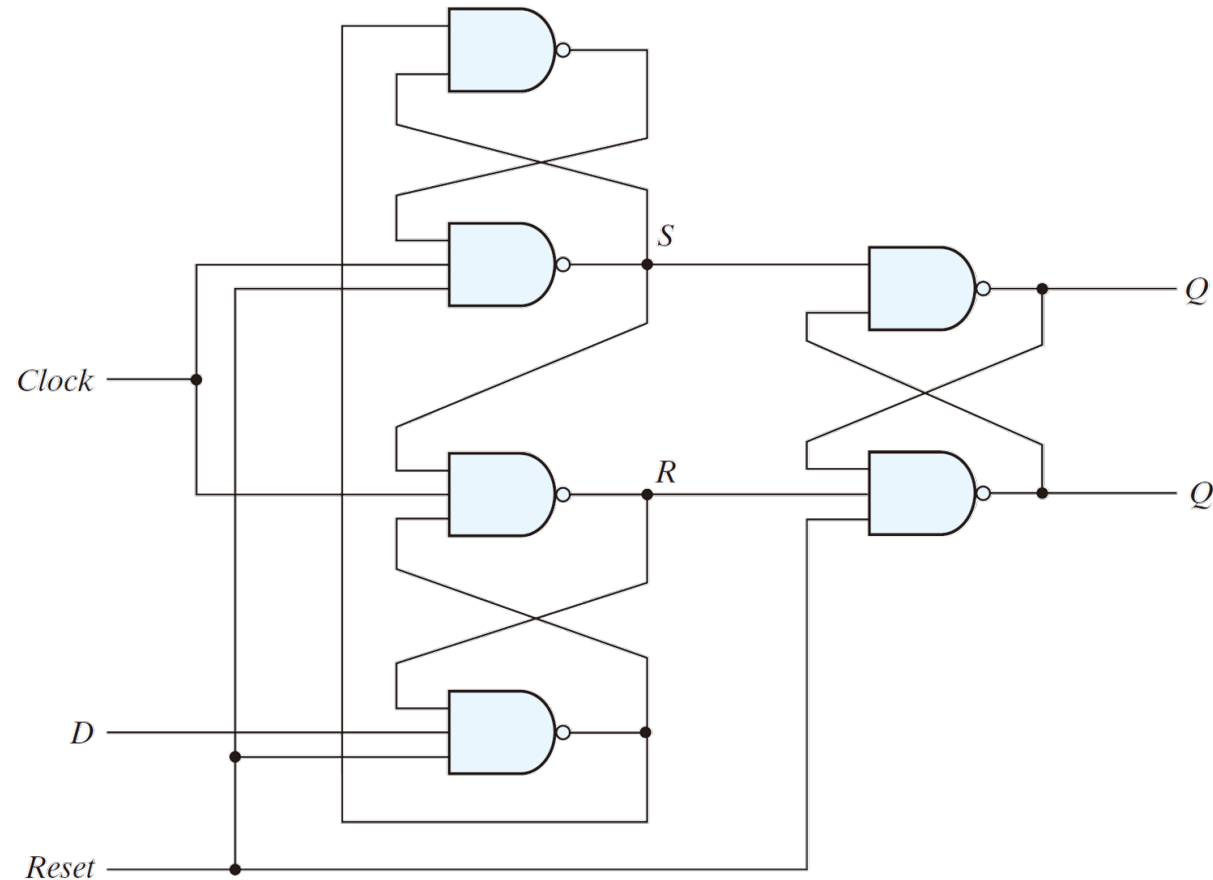
Other Flip-Flops

- Characteristic Equations (特徵方程式、特性方程式)
 - D flip-flop
 - $Q(t + 1) = D$
 - JK flip-flop
 - $Q(t + 1) = JQ' + K'Q$
 - T flop-flop
 - $Q(t + 1) = T \oplus Q$

5.4 Storage Element: Flip-Flops (14/16)

Direct Inputs

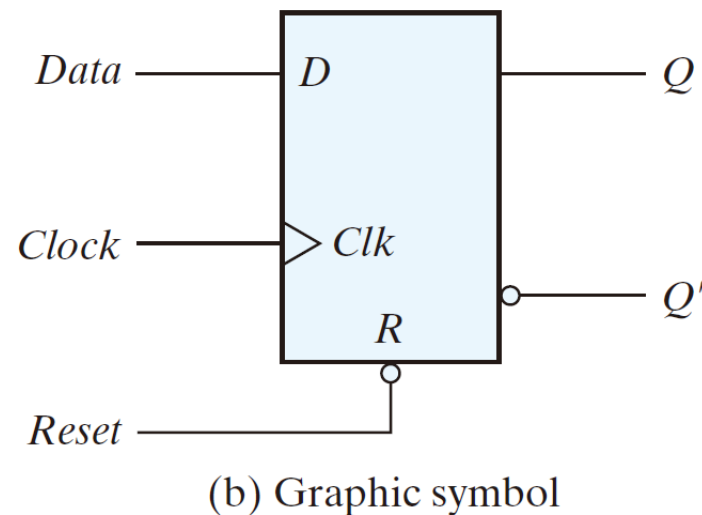
- Asynchronous *Set* and/or asynchronous *Reset*



(a) Circuit diagram

5.4 Storage Element: Flip-Flops (15/16)

Direct Inputs



R	Clk	D	Q	Q'
0	X	X	0	1
1	$\uparrow$	0	0	1
1	$\uparrow$	1	1	0

(b) Function table

FIGURE 5.14
D flip-flop with asynchronous reset

5.4 Storage Element: Flip-Flops (16/16)

Direct Inputs

- **Practice Exercise 5.5** Describe the functionality of a *D*-type flip-flop.

Answer:

1. A *D*-type flip-flop has a *D* (data) input, a clock input, and possibly asynchronous or synchronous *reset* or *set* signal.
2. If *set* or *reset* are not asserted, the clock signal synchronizes the transfer of *D* to the output *Q*.
3. If *set* or *reset* are asynchronous, their action controls the flip-flop independently of the clock. *set* causes the output to be 1; *reset* causes the output to be 0.
4. If *set* or *reset* are synchronous, their action has effect at the synchronizing edge of the clock.

5.5 Analysis of Clocked Sequential Circuits (1/33)

State Equations

- A sequential circuit
 - (inputs, current state) $\Rightarrow$ (output, next state)
 - state transition table or **state table**
 - state transition diagram or **state diagram**
- A **state equation** (also called a transition equation) specifies the next state of the sequential circuit as a function of the present state and input.
- An **output equation** specifies the output of the sequential circuit as a function of the present state and input.
- Design steps
 - Circuit diagram $\rightarrow$ Equations $\rightarrow$ State table $\rightarrow$ State diagram
 - State diagram $\rightarrow$ State table $\rightarrow$ Equations $\rightarrow$ Circuit diagram

5.5 Analysis of Clocked Sequential Circuits (2/33)

State Equations

Circuit → **Equations** → State table → State diagram

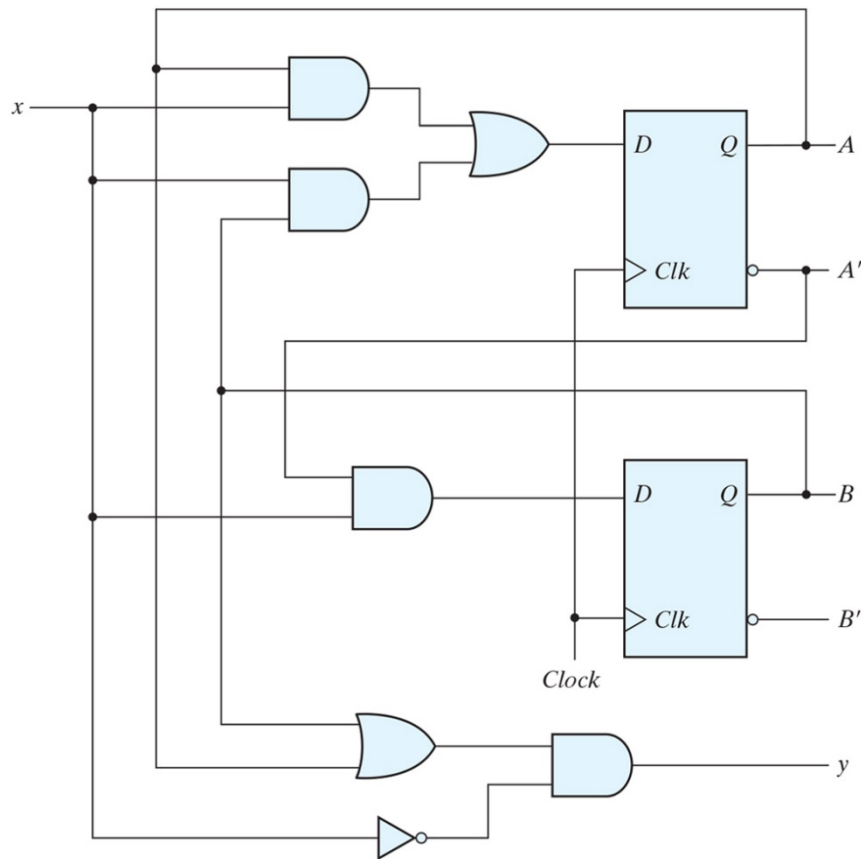


FIGURE 5.15
Example of sequential circuit

- The state equations
 - $A(t + 1) = A(t)x(t) + B(t)x(t)$
 - $B(t + 1) = A'(t)x(t)$
- A more compact form
 - $A(t + 1) = Ax + Bx$
 - $B(t + 1) = A'x$
- The output equation
 - $y(t) = [A(t) + B(t)]x'(t)$
 - $y = (A + B)x'$

5.5 Analysis of Clocked Sequential Circuits (3/33)

State Table

Circuit → Equations → State table → State diagram

- The state equations
 - $A(t + 1) = A(t)x(t) + B(t)x(t)$
 - $B(t + 1) = A'(t)x(t)$
- A more compact form
 - $A(t + 1) = Ax + Bx$
 - $B(t + 1) = A'x$
- The output equation
 - $y(t) = [A(t) + B(t)]x'(t)$
 - $y = (A + B)x'$

Table 5.2
State Table for the Circuit of Fig. 5.15

Present State		Input	Next State		Output
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

5.5 Analysis of Clocked Sequential Circuits (4/33)

State Diagram

Circuit → Equations → State table → State diagram

- A circle: a state
- A directed line: the transition between the circles (or states).
 - Each directed line is labeled 'inputs/outputs'
 - Each line originates at a present state and terminates at a next state, depending on the input applied when the circuit is in the present state.

Table 5.3
Second Form of the State Table

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

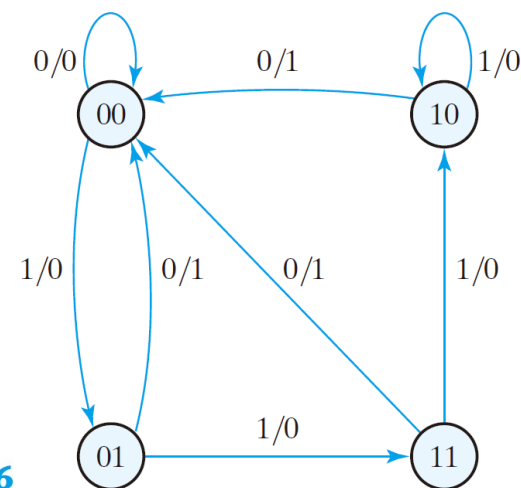


FIGURE 5.16
State diagram of the circuit of Fig. 5.15

5.5 Analysis of Clocked Sequential Circuits (5/33)

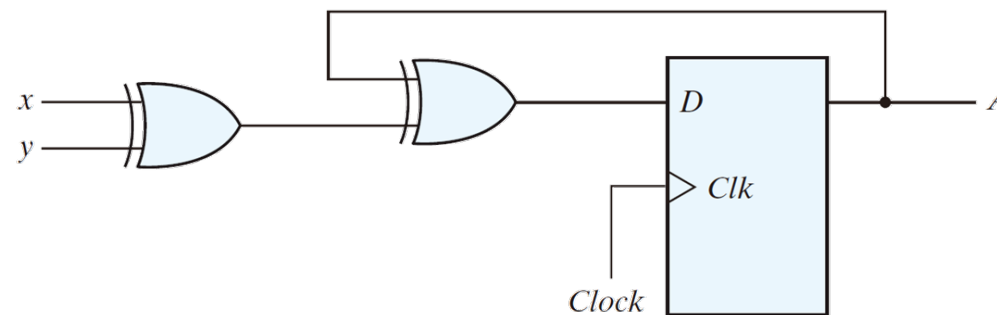
Flip-Flop Input Equations

- A sequential circuit can be fully described by the input (flip-flop-related) and output (logic gates-related) equations.
- Input equation: a set of Boolean functions to generate the inputs to the flip-flops.
 - Also called excitation functions (激勵方程式)
 - $D_A = Ax + Bx$
 - $D_B = A'x$
 - The symbols in input equations, D_A and D_B , imply the type of the flip-flop from the letter symbol
- Output equations: a set of Boolean functions to generate external outputs.
 - $y = (A + B)x'$

5.5 Analysis of Clocked Sequential Circuits (6/33)

Analysis with *D* Flip-Flops

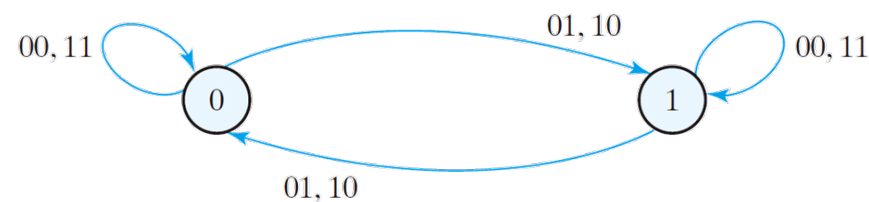
- The input equation
 - $D_A = A \oplus x \oplus y$
- The state equation
 - $A(t + 1) = A \oplus x \oplus y$
- Note that the input equation for *D* flip-flop is identical to the corresponding state equation.



(a) Circuit diagram

Present state	Inputs		Next state
<i>A</i>	<i>x</i>	<i>y</i>	<i>A</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

(b) State table



(c) State diagram

FIGURE 5.17
Sequential circuit with *D* flip-flop

5.5 Analysis of Clocked Sequential Circuits (7/33)

Analysis with D Flip-Flops

- **Practice Exercise 5.6** What determines the next state of a D -type flip-flop?

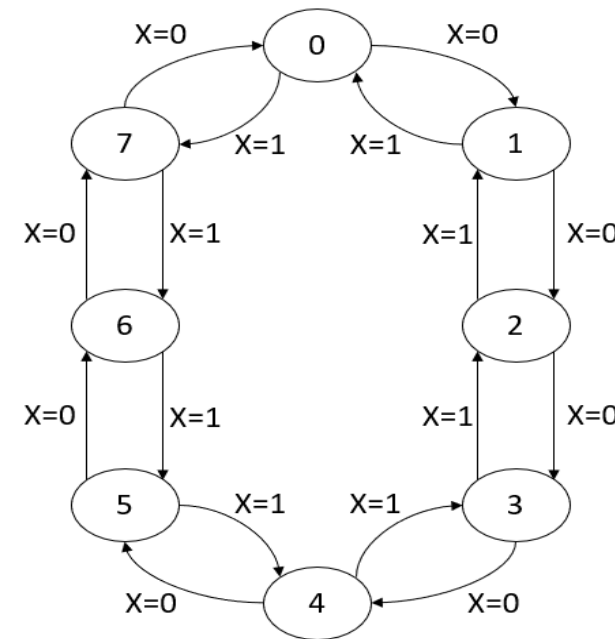
Answer: The next state of a D -type flip-flop is the value of D at the synchronizing edge of the clock.

5.5 Analysis of Clocked Sequential Circuits (8/33)

Analysis with *D* Flip-Flops: Example 1

Problem → **State diagram** → State table → Equations → Circuit

- 設計具有3個*D*型正反器*A*, *B*, *C*與1個輸入*X*之3-bit counter。
- $X=0$ 時，電路會上數
- $X=1$ 時，電路會下數

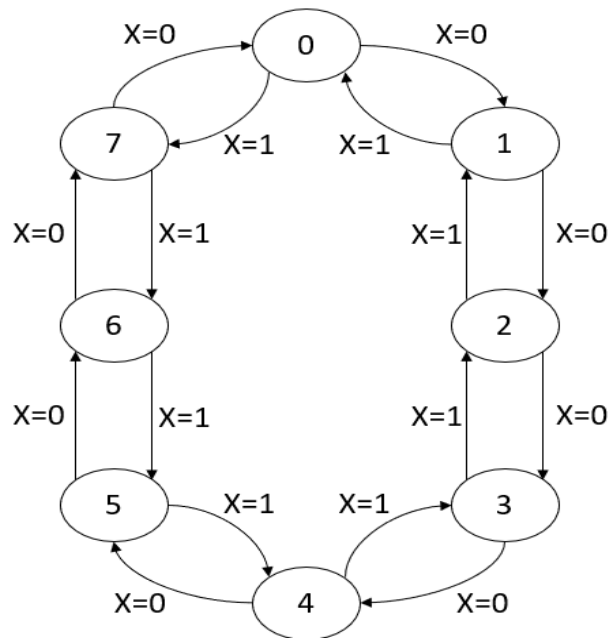


State diagram

5.5 Analysis of Clocked Sequential Circuits (9/33)

Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit



State diagram

Present State			Input <i>X</i>	Next State		
<i>A</i>	<i>B</i>	<i>C</i>		<i>A</i>	<i>B</i>	<i>C</i>
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

5.5 Analysis of Clocked Sequential Circuits (10/33)

Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State		
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

<i>AB</i> \ <i>CX</i>	00	01	11	10
00		1		
01				1
11	1	1	1	
10	1		1	1

$$D_A = AC'X + A'B'C'X + ABX + AB'C + A'BCX'$$

5.5 Analysis of Clocked Sequential Circuits (11/33)

Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State		
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

<i>AB</i> \ <i>CX</i>	00	01	11	10
00		1		1
01	1		1	
11	1		1	
10		1		1

$$D_B = BC'X + B'C'X + BCX + B'CX$$

5.5 Analysis of Clocked Sequential Circuits (12/33)

Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State		
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>
0	0	0	0	0	0	1
0	0	0	1	1	1	1
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	1	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	1
1	0	1	0	1	1	0
1	0	1	1	1	0	0
1	1	0	0	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0
1	1	1	1	1	1	0

State table

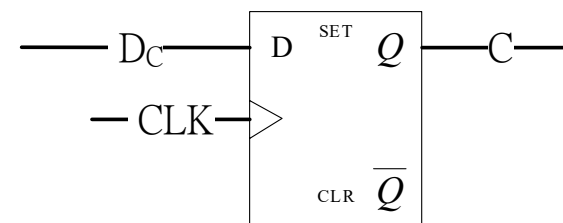
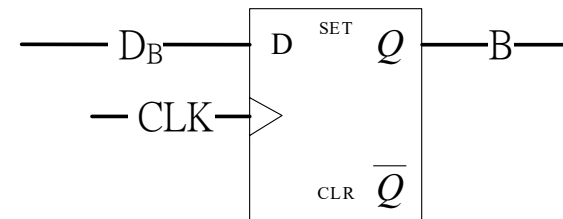
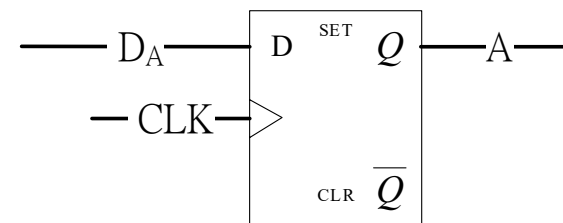
<i>AB</i> \ <i>CX</i>	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1		

$$D_C = C'$$

5.5 Analysis of Clocked Sequential Circuits (13/33)

Analysis with *D* Flip-Flops: Example 1

Problem → State diagram → State table → Equations → Circuit

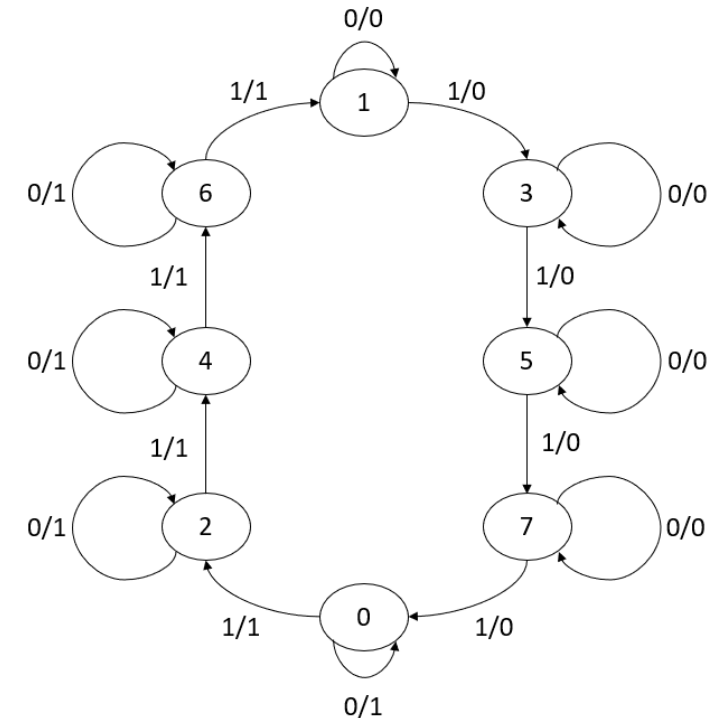


5.5 Analysis of Clocked Sequential Circuits (14/33)

Analysis with *D* Flip-Flops: Example 2

Problem → **State diagram** → State table → Equations → **Circuit**

- 設計具有3個*D*型正反器*A, B, C*、1個輸入*X*與1個輸出*Y*之3-bit counter。
 - $X=0$ 時，狀態值保持不變
 - $X=1$ 時，
 - 目前狀態值為奇數時，
 - 若目前狀態值為7，次一狀態值為0。
 - 否則，狀態值加2。
 - 目前狀態值為偶數時，
 - 若目前狀態值為6，次一狀態值為1。
 - 否則，狀態值加2。
 - 目前狀態值為奇數時， $Y=0$ ，為偶數時， $Y=1$

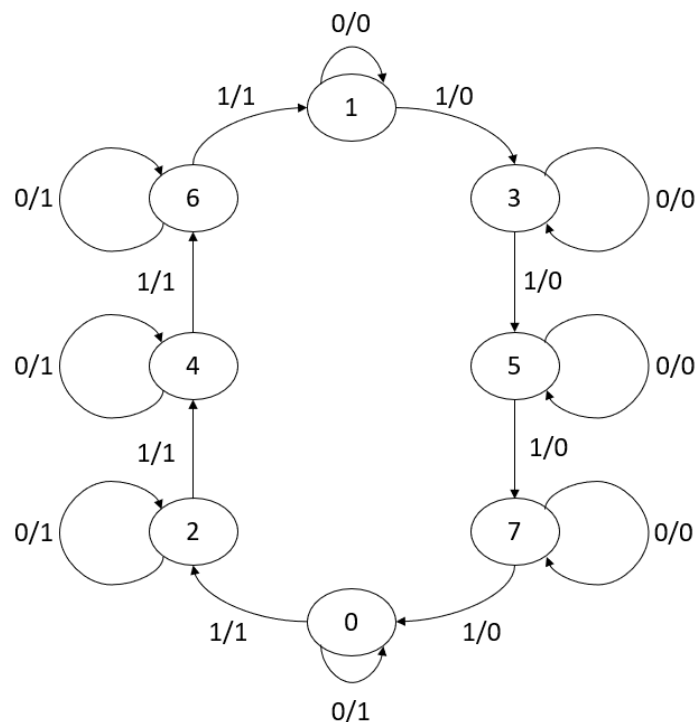


State diagram

5.5 Analysis of Clocked Sequential Circuits (15/33)

Analysis with *D* Flip-Flops: Example 2

Problem → State diagram → State table → Equations → Circuit



State diagram

Present State			Input	Next State			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>Y</i>
0	0	0	0	0	0	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	1	1	0
0	1	0	0	0	1	0	1
0	1	0	1	1	0	0	1
0	1	1	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	1	0	0	1
1	0	0	1	1	1	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	1	1	0
1	1	0	0	1	1	0	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	1	0
1	1	1	1	0	0	0	0

State table

5.5 Analysis of Clocked Sequential Circuits (16/33)

Analysis with *D* Flip-Flops: Example 2

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>Y</i>
0	0	0	0	0	0	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	1	1	0
0	1	0	0	0	1	0	1
0	1	0	1	1	0	0	1
0	1	1	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	1	0	0	1
1	0	0	1	1	1	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	1	1	0
1	1	0	0	1	1	0	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	1	0
1	1	1	1	0	0	0	0

State table

<i>AB</i> \ <i>CX</i>	00	01	11	10
00				
01		1	1	
11	1			1
10	1	1	1	1

$$D_A = AC'X' + A'BX + AB'X + ACX'$$

5.5 Analysis of Clocked Sequential Circuits (17/33)

Analysis with *D* Flip-Flops: Example 2

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>Y</i>
0	0	0	0	0	0	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	1	1	0
0	1	0	0	0	1	0	1
0	1	0	1	1	0	0	1
0	1	1	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	1	0	0	1
1	0	0	1	1	1	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	1	1	0
1	1	0	0	1	1	0	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	1	0
1	1	1	1	0	0	0	0

State table

<i>AB</i> \ <i>CX</i>	00	01	11	10
00		1	1	
01	1			1
11	1			1
10		1	1	

$$D_B = BC'X' + A'B'X + AB'X + BCX'$$

5.5 Analysis of Clocked Sequential Circuits (18/33)

Analysis with *D* Flip-Flops: Example 2

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>Y</i>
0	0	0	0	0	0	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	1	1	0
0	1	0	0	0	1	0	1
0	1	0	1	1	0	0	1
0	1	1	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	1	0	0	1
1	0	0	1	1	1	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	1	1	0
1	1	0	0	1	1	0	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	1	0
1	1	1	1	0	0	0	0

State table

<i>AB</i> \ <i>CX</i>	00	01	11	10
00			1	1
01			1	1
11		1		1
10			1	1

$$D_C = ABC'X + A'C + B'C + CX'$$

5.5 Analysis of Clocked Sequential Circuits (19/33)

Analysis with *D* Flip-Flops: Example 2

Problem → State diagram → State table → Equations → Circuit

Present State			Input	Next State			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>Y</i>
0	0	0	0	0	0	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	0	1	0
0	0	1	1	0	1	1	0
0	1	0	0	0	1	0	1
0	1	0	1	1	0	0	1
0	1	1	0	0	1	1	0
0	1	1	1	1	0	1	0
1	0	0	0	1	0	0	1
1	0	0	1	1	1	0	1
1	0	1	0	1	0	1	0
1	0	1	1	1	1	1	0
1	1	0	0	1	1	0	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	1	0
1	1	1	1	0	0	0	0

State table

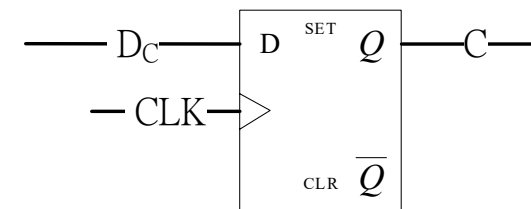
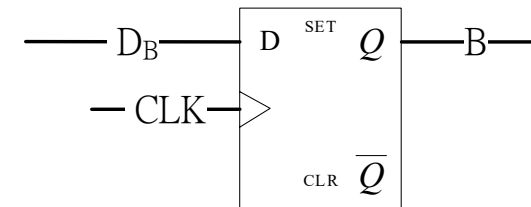
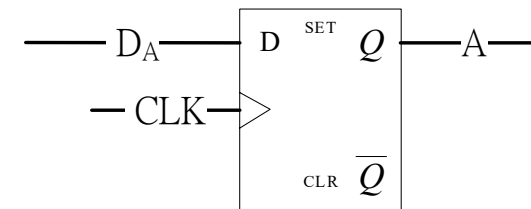
<i>AB</i> \ <i>CX</i>	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1		

$$D_C = C'$$

5.5 Analysis of Clocked Sequential Circuits (20/33)

Analysis with *D* Flip-Flops: Example 2

Problem → State diagram → State table → Equations → Circuit



Y

5.5 Analysis of Clocked Sequential Circuits (21/33)

Analysis with *JK* Flip-Flops

- Procedures to derive the next-state of a sequential circuit that uses *JK*- or *T*-type flip-flops.
 - Determine the flip-flop input equations in terms of the present state and inputs.
 - List the binary values of each input equations.
 - Use the corresponding flip-flop characteristic table to determine the next-state values in the state table.

Table 5.1
Flip-Flop Characteristic Tables

<i>JK</i> Flip-Flop $Q(t + 1) = JQ' + K'Q$			
<i>J</i>	<i>K</i>	$Q(t + 1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

<i>D</i> Flip-Flop $Q(t + 1) = D$		
<i>D</i>	$Q(t + 1)$	
0	0	Reset
1	1	Set

<i>T</i> Flip-Flop $Q(t + 1) = T \oplus Q$		
<i>T</i>	$Q(t + 1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

5.5 Analysis of Clocked Sequential Circuits (22/33)

Analysis with *JK* Flip-Flops

Circuit → **Equations** → State table → State diagram

• Circuit

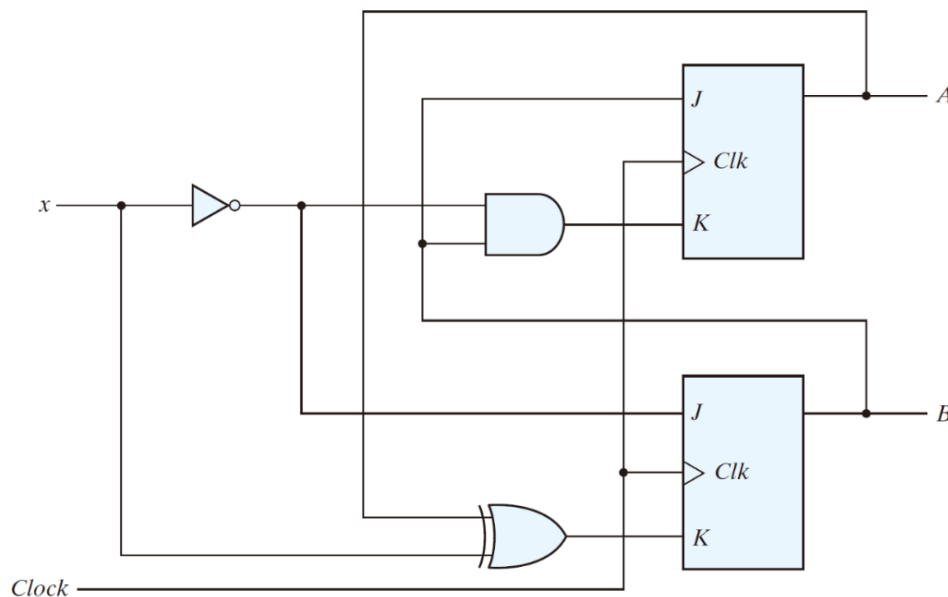


FIGURE 5.18
Sequential circuit with *JK* flip-flop

• Input equations

- $J_A = B, K_A = Bx'$
- $J_B = x', K_B = A'x + Ax' = A \oplus x$

• State equations

- Characteristic equations of *JK* flip-flop.
 - $A(t+1) = J_A A' + K_A' A$
 - $B(t+1) = J_B B' + K_B' B$
- $A(t+1) = BA' + (Bx')'A$
 $= A'B + AB' + Ax$
- $B(t+1) = x'B' + (A \oplus x)'B$
 $= B'x' + ABx + A'Bx'$

5.5 Analysis of Clocked Sequential Circuits (23/33)

Analysis with *JK* Flip-Flops

Circuit → Equations → State table → State diagram

• Input equations

- $J_A = B, K_A = Bx'$
- $J_B = x', K_B = A'x + Ax' = A \oplus x$

• State equations

- Derive the state equations by using characteristic equations of *JK* flip-flop.
 - $A(t+1) = J_A A' + K_A A$
 - $B(t+1) = J_B B' + K_B B$
- $A(t+1) = BA' + (Bx')'A$
 $= A'B + AB' + Ax$
- $B(t+1) = x'B' + (A \oplus x)'B$
 $= B'x' + ABx + A'Bx'$

• State table

Table 5.4
State Table for Sequential Circuit with *JK* Flip-Flops

Present State		Input	Next State	
<i>A</i>	<i>B</i>		<i>A</i>	<i>B</i>
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

5.5 Analysis of Clocked Sequential Circuits (24/33)

Analysis with *JK* Flip-Flops

Circuit $\rightarrow$ Equations $\rightarrow$ State table $\rightarrow$ State diagram

- State table
- State transition diagram

Table 5.4
State Table for Sequential Circuit with JK Flip-Flops

Present State		Input	Next State	
<i>A</i>	<i>B</i>		<i>A</i>	<i>B</i>
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

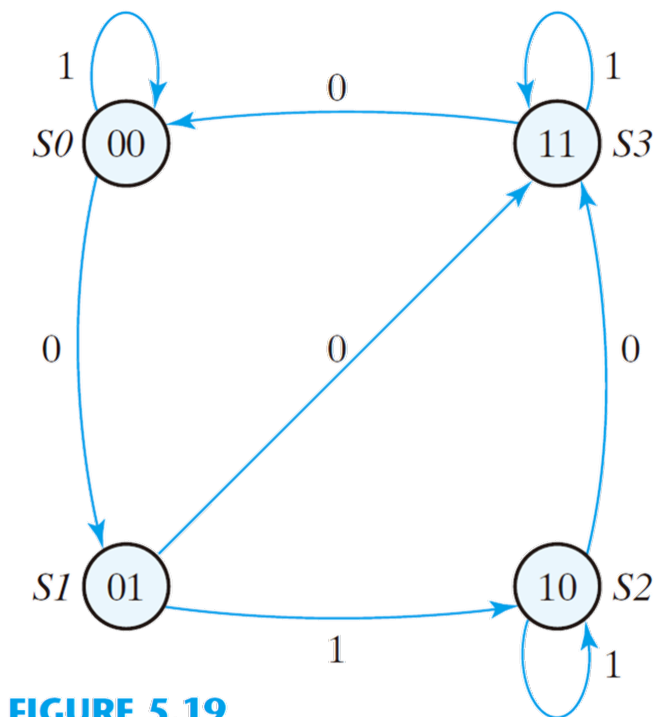


FIGURE 5.19
State diagram of the circuit of Fig. 5.18

5.5 Analysis of Clocked Sequential Circuits (25/33)

Analysis with JK Flip-Flops

- **Practice Exercise 5.7** What determines the next state of a JK -type flip-flop?

Answer: The next state of a JK -type flip-flop is determined by the value of inputs J and K at the synchronizing edge of the clock.

5.5 Analysis of Clocked Sequential Circuits (26/33)

Analysis with T Flip-Flops

Circuit $\rightarrow$ Equations $\rightarrow$ State table $\rightarrow$ State diagram

• Circuit

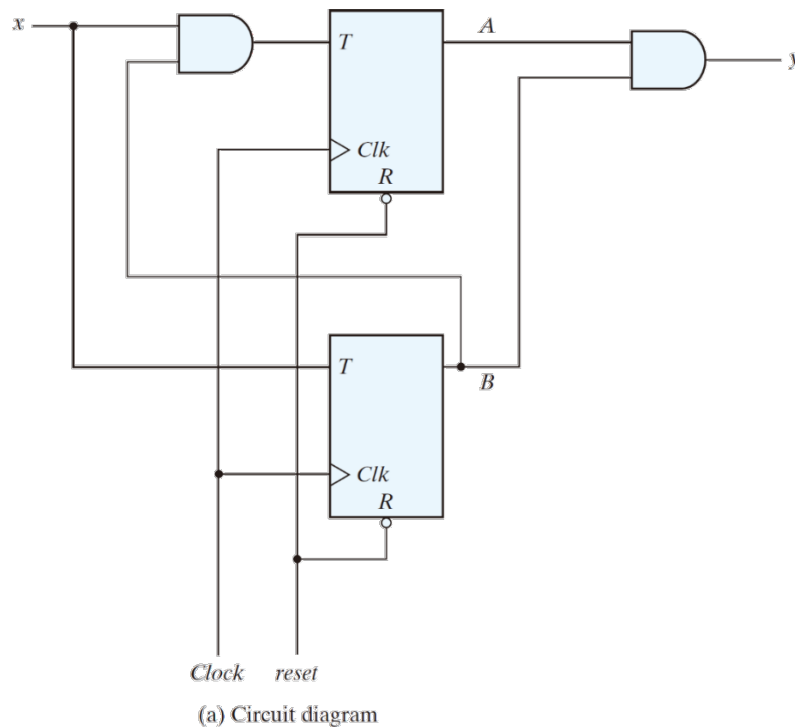


FIGURE 5.20
Sequential circuit with T flip-flops (Binary Counter)

• The input and output functions

- $T_A = Bx, T_B = x$
- $y = AB$
 - 輸出 y 只有與目前狀態 AB 有關與輸入 x 無關

• The state equations

- Characteristic equations of T flip-flop
 - $A(t+1) = T_A \oplus A = T_A'A + T_AA'$
 - $B(t+1) = T_B \oplus B = T_B'B + T_BB'$
- $A(t+1) = (Bx)'A + (Bx)A'$
 $= AB' + Ax' + A'Bx$
- $B(t+1) = x \oplus B$

5.5 Analysis of Clocked Sequential Circuits (27/33)

Analysis with T Flip-Flops

Circuit $\rightarrow$ Equations $\rightarrow$ State table $\rightarrow$ State diagram

- The input and output functions

- $T_A = Bx, T_B = x$

- $y = AB$

- 輸出 y 只有與目前狀態 AB 有關與輸入 x 無關

- The state equations

- Derive the state equations by using characteristic equations of T flip-flop.

- $Q(t+1) = T \oplus Q = T'Q + TQ'$

- $A(t+1) = (Bx)'A + (Bx)A'$
 $= AB' + Ax' + A'Bx$

- $B(t+1) = x \oplus B$

- State table

Table 5.5
State Table for Sequential Circuit with T Flip-Flops

Present State		Input	Next State		Output
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

5.5 Analysis of Clocked Sequential Circuits (28/33)

Analysis with *T* Flip-Flops

Circuit → Equations → State table → State diagram

- State table

Table 5.5
State Table for Sequential Circuit with T Flip-Flops

Present State		Input <i>x</i>	Next State		Output <i>y</i>
<i>A</i>	<i>B</i>		<i>A</i>	<i>B</i>	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

- State transition diagram

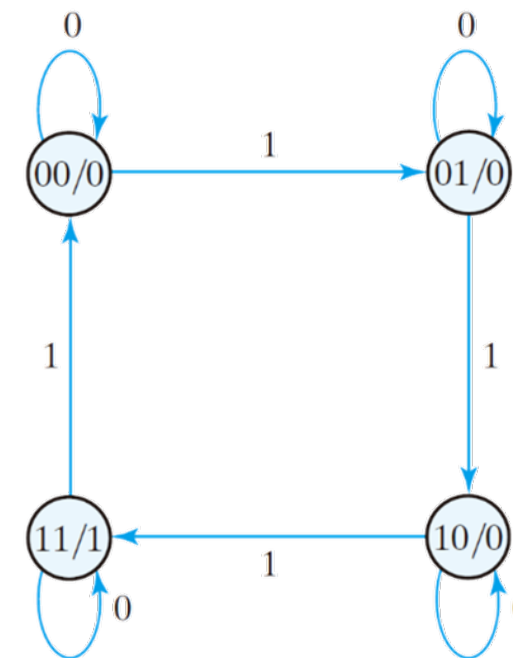


FIGURE 5.20 (b) State diagram
Sequential circuit with *T* flip-flops (Binary Counter)

5.5 Analysis of Clocked Sequential Circuits (29/33)

Mealy and Moore Models of Finite State Machines

- A sequential circuit can be classified into two models:
 - The Mealy model: the outputs are functions of both the present state and inputs (Fig. 5.15)
 - The outputs may change if the inputs change during the clock pulse period
 - The outputs may have momentary false values unless the inputs are synchronized with the clocks
 - The Moore model: the outputs are functions of the present state only (Fig. 5.20)
 - The outputs are synchronous with the clocks
- The two models of sequential circuits are commonly referred to as a finite state machine (FSM).
 - Mealy machine or Mealy FSM
 - Moore machine or Moore FSM

5.5 Analysis of Clocked Sequential Circuits (30//33)

Mealy and Moore Models of Finite State Machines

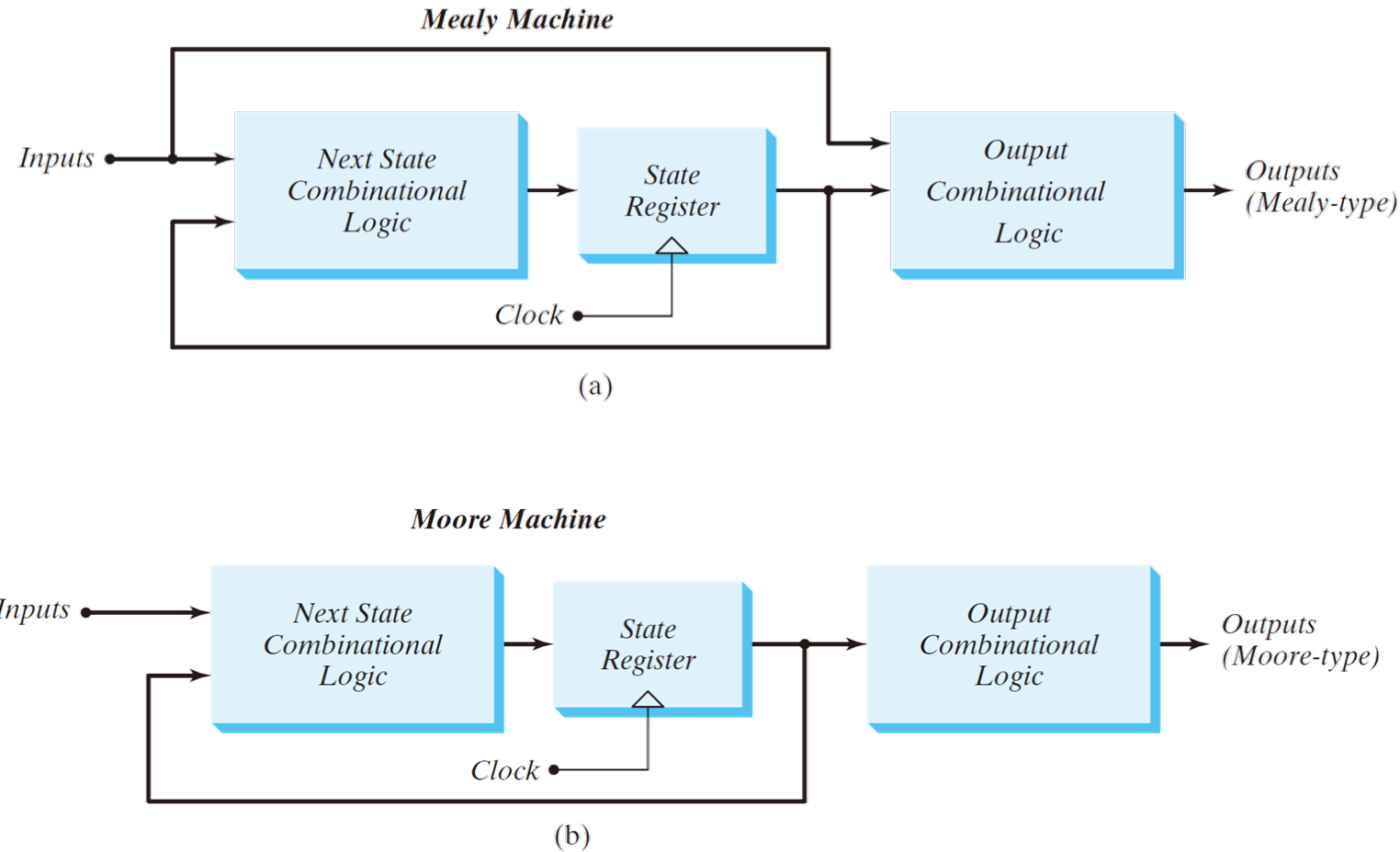


FIGURE 5.21

Block diagrams of Mealy and Moore state machines

5.5 Analysis of Clocked Sequential Circuits (31/33)

Mealy and Moore Models of Finite State Machines

- **Practice Exercise 5.8** What determines the next state of a T -type flip-flop?

Answer: If the T input is asserted, the next state is the complement of the present state (output) at the synchronizing edge of the clock. If T is not asserted, the state remains fixed.

- **Practice Exercise 5.9** What is the difference between a Mealy and Moore state machine?

Answer: The output of a Moore state machine depends on only the state of the machine, while the output of a Mealy machine depends on the present state and the inputs to the machine.

5.5 Analysis of Clocked Sequential Circuits (32/33)

Mealy and Moore Models of Finite State Machines

- **Practice Exercise 5.10** What does an edge of a state machine chart represent?

Answer: The edge of a state machine chart represents a transition of the machine between two states.

- **Practice Exercise 5.11** In a synchronous finite state machine, when does a transition between states occur?

Answer: A transition between the states of a finite state machine occurs at the active edge of the synchronizing signal (clock).

- **Practice Exercise 5.12** What kinds of reset may a finite state machine have?

Answer: A finite state machine may have synchronous or asynchronous reset.

5.5 Analysis of Clocked Sequential Circuits (33/33)

Mealy and Moore Models of Finite State Machines

- **Practice Exercise 5.13** Cite a reason why it is an important practice to implement a reset signal in a finite state machine?

Answer: Without a reset signal, a finite state machine cannot be driven into a known initial state.

- **Practice Exercise 5.14** What type of finite state machine may have an output that depends on one or more inputs?

Answer: The outputs of a Mealy state machine may depend on the inputs to the machine.

5.7 State Reduction and Assignment (1/9)

State Reduction

- State-reduction algorithms are concerned with procedures for reducing the number of states in a state table, while keeping the external input-output requirements unchanged.
- A reduction in the number of states may result in a reduction in the number of flip-flops and the number of gates.

- An example state diagram

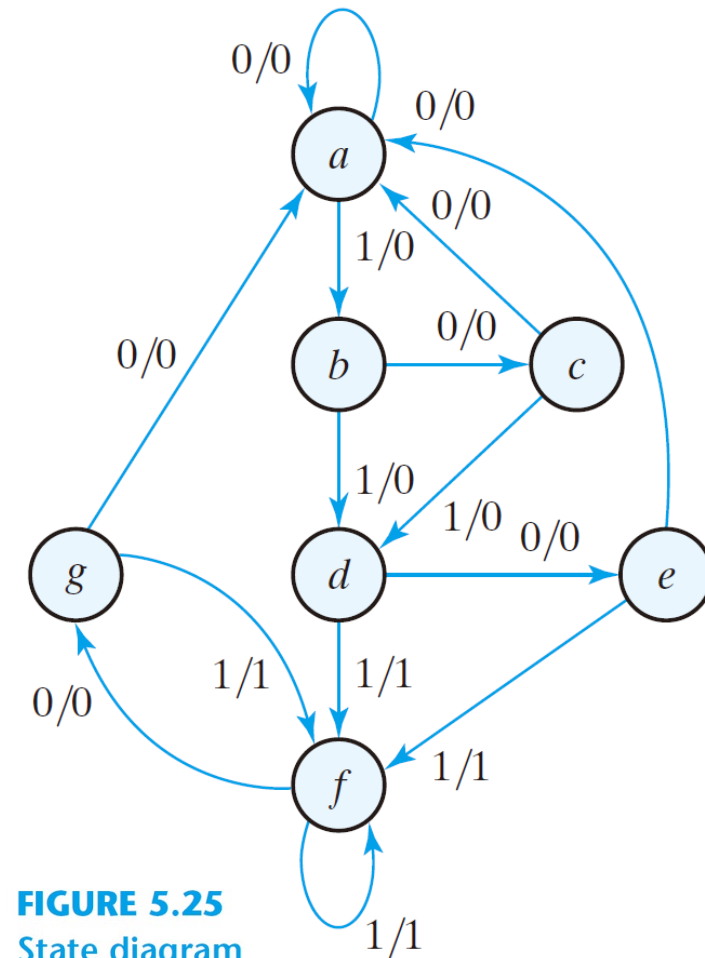


FIGURE 5.25
State diagram

5.7 State Reduction and Assignment (2/9)

State Reduction

state	<i>a</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>f</i>	<i>g</i>	<i>f</i>	<i>g</i>	<i>a</i>
input	0	1	0	1	0	1	1	0	1	0	0	
output	0	0	0	0	0	1	1	0	1	0	0	

- Only the input-output sequences are important
- Two circuits are equivalent
 - have identical outputs for all input sequences
 - the number of states is not important

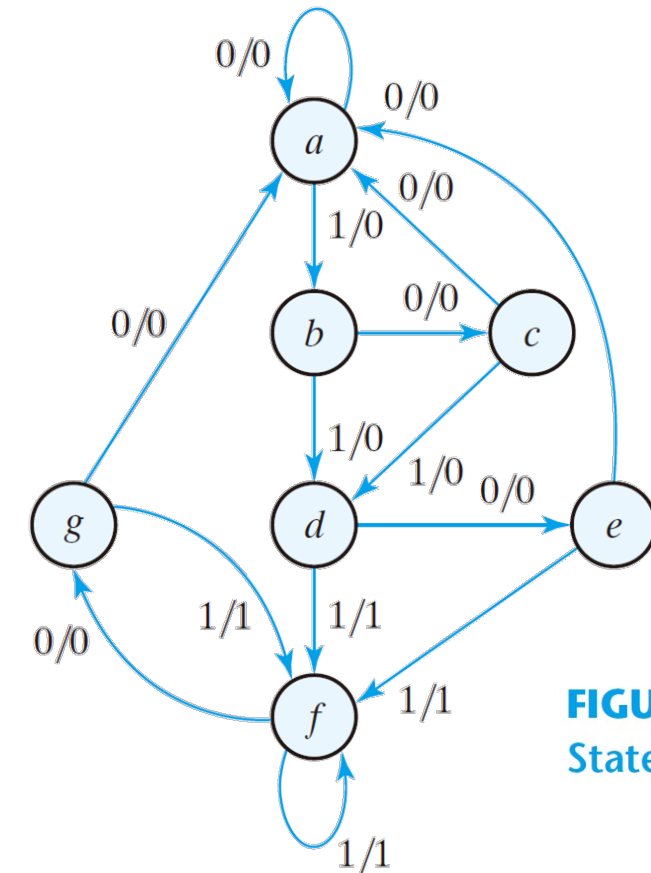


FIGURE 5.25
State diagram

5.7 State Reduction and Assignment (3/9)

State Reduction

- Two states are said to be equivalent
 - If their present states will go to the same next state and have the same output for both input combinations.
 - One of them can be removed.

Table 5.6
State Table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
<i>a</i>	<i>a</i>	<i>b</i>	0	0
<i>b</i>	<i>c</i>	<i>d</i>	0	0
<i>c</i>	<i>a</i>	<i>d</i>	0	0
<i>d</i>	<i>e</i>	<i>f</i>	0	1
<i>e</i>	<i>a</i>	<i>f</i>	0	1
<i>f</i>	<i>g</i>	<i>f</i>	0	1
<i>g</i>	<i>a</i>	<i>f</i>	0	1

5.7 State Reduction and Assignment (4/9)

State Reduction

- States e and g are equivalent states.
- Removing state g .
- Replacing state g with state e .

Table 5.6
State Table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

Table 5.7
Reducing the State Table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	e	f	0	1

5.7 State Reduction and Assignment (5/9)

State Reduction

- States d and f are equivalent states.
- Removing state f .
- Replacing state f with state d .

Table 5.7
Reducing the State Table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	e	f	0	1

Table 5.8
Reduced State Table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

5.7 State Reduction and Assignment (6/9)

State Reduction

- The reduced finite state machine

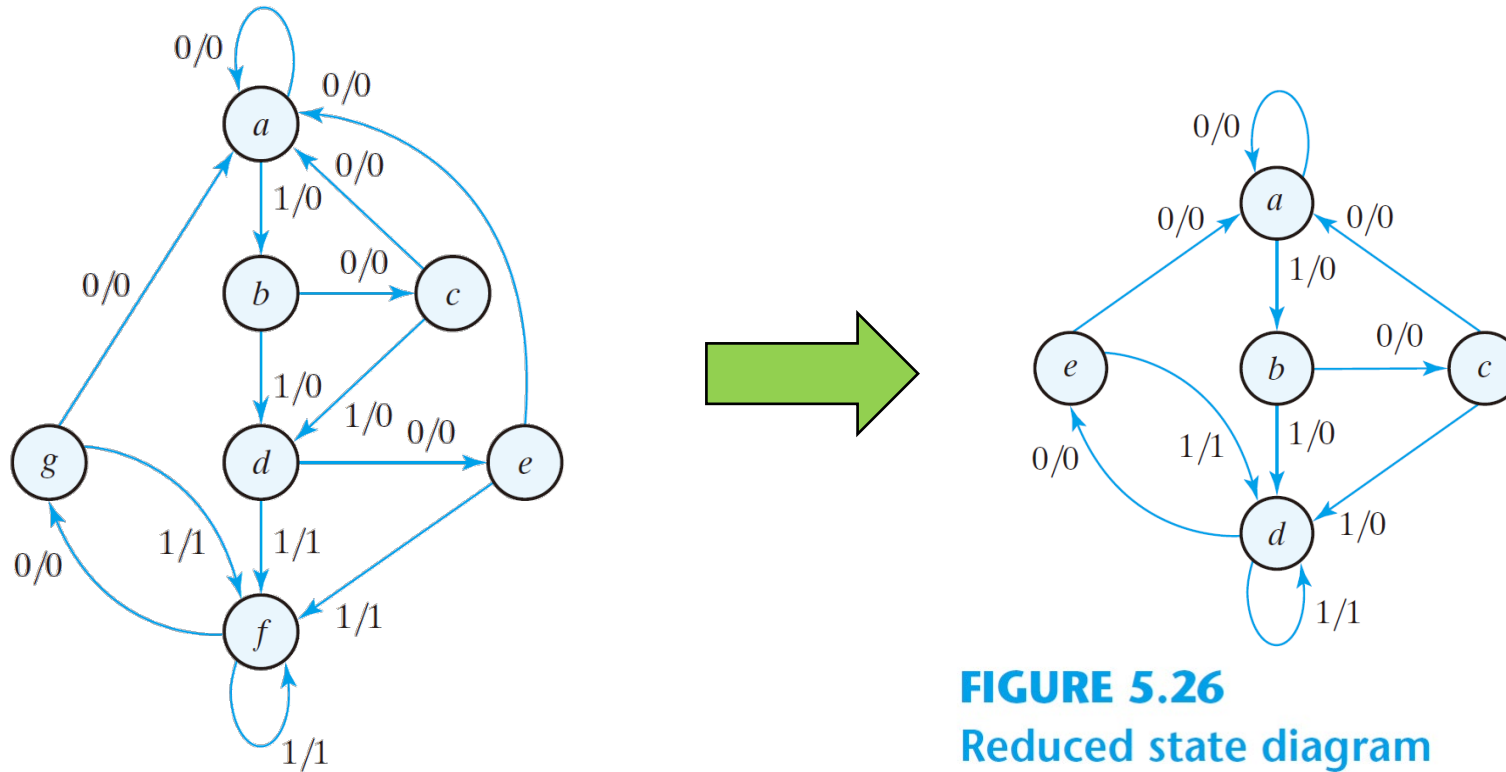
Table 5.8
Reduced State Table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
<i>a</i>	<i>a</i>	<i>b</i>	0	0
<i>b</i>	<i>c</i>	<i>d</i>	0	0
<i>c</i>	<i>a</i>	<i>d</i>	0	0
<i>d</i>	<i>e</i>	<i>d</i>	0	1
<i>e</i>	<i>a</i>	<i>d</i>	0	1

OLD	state	<i>a</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>f</i>	<i>g</i>	<i>f</i>	<i>g</i>	<i>a</i>
	input	0	1	0	1	0	1	1	0	1	0	0	
	output	0	0	0	0	0	1	1	0	1	0	0	
NEW	state	<i>a</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>d</i>	<i>d</i>	<i>e</i>	<i>d</i>	<i>e</i>	<i>a</i>
	input	0	1	0	1	0	1	1	0	1	0	0	
	output	0	0	0	0	0	1	1	0	1	0	0	

5.7 State Reduction and Assignment (7/9)

State Reduction



5.7 State Reduction and Assignment (8/9)

State Assignment

- Three possible binary state assignments
- The unused states are treated as don't-care condition $\Rightarrow$ fewer combinational gates

Table 5.9
Three Possible Binary State Assignments

State	Assignment 1, Binary	Assignment 2, Gray Code	Assignment 3, One-Hot
<i>a</i>	000	000	00001
<i>b</i>	001	001	00010
<i>c</i>	010	011	00100
<i>d</i>	011	010	01000
<i>e</i>	100	110	10000

5.7 State Reduction and Assignment (9/9)

State Assignment

- Use binary assignment 1

Table 5.10
Reduced State Table with Binary Assignment 1

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
000	000	001	0	0
001	010	011	0	0
010	000	011	0	0
011	100	011	0	1
100	000	011	0	1

5.8 Design Procedure (1/28)

- The word description of the circuit behavior
- Draw the corresponding state diagram
- State reduction if necessary
- Assign binary values to the states
- Obtain the binary-coded state table
- Choose the type of flip-flops
- Derive the simplified flip-flop input equations and output equations
- Draw the logic diagram

5.8 Design Procedure (2/28)

Sequence Detector --- Synthesis Using *D* Flip-Flops

Problem → **State diagram** → State table → Equations → Circuit

- An example state diagram and state table, Moore's machine
 - If the input sequence "111" is detected, output $y = 1$. Otherwise, output $y = 0$.
 - Input x : 1111010...
 - Output y : 0011000

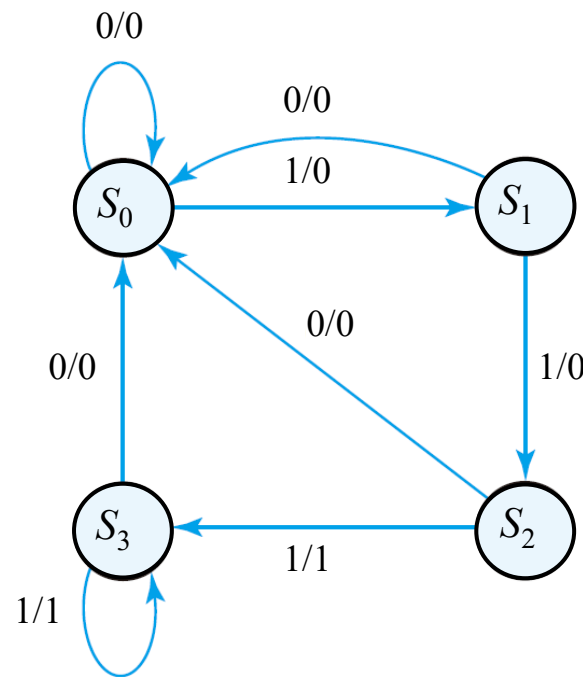


FIGURE 5.27

State diagram for sequence detector

5.8 Design Procedure (3/28)

Sequence Detector --- Synthesis Using D Flip-Flops

Problem → State diagram → State table → Equations → Circuit

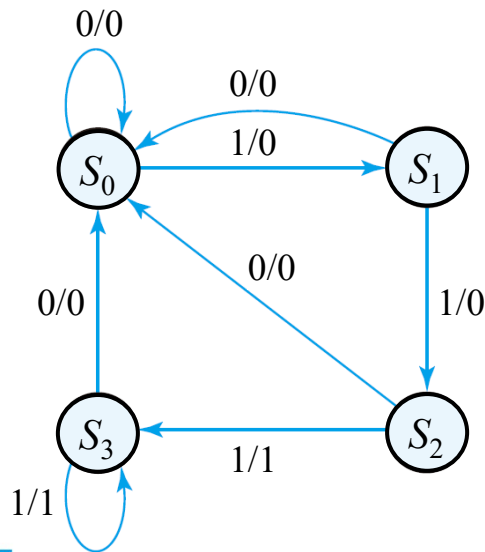


FIGURE 5.27
State diagram for sequence detector

Table 5.11
State Table for Sequence Detector

Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

5.8 Design Procedure (4/28)

Sequence Detector --- Synthesis Using *D* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

Table 5.11
State Table for Sequence Detector

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

- The output equation

- $y(A, B) = \Sigma(6, 7)$

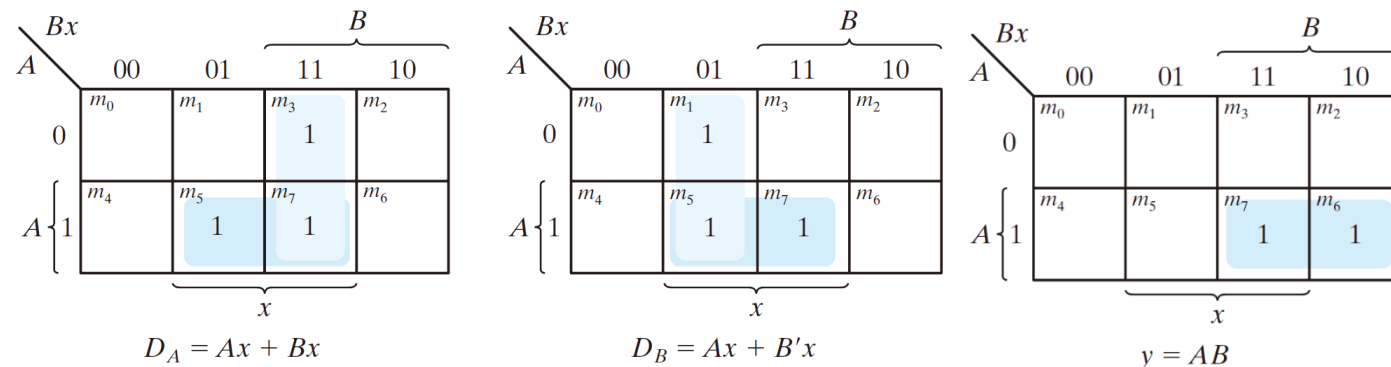


FIGURE 5.28
K-Maps for sequence detector

- The flip-flop input equations

- $A(t + 1) = D_A(A, B, x) = \Sigma(3, 5, 7)$
- $B(t + 1) = D_B(A, B, x) = \Sigma(1, 5, 7)$

- Logic minimization using the K-map

- $D_A = Ax + Bx$
- $D_B = Ax + B'x$
- $y = AB$

5.8 Design Procedure (5/28)

Sequence Detector --- Synthesis Using *D* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

- Logic minimization using the K-map

- $D_A = Ax + Bx$
- $D_B = Ax + B'x$
- $y = AB$

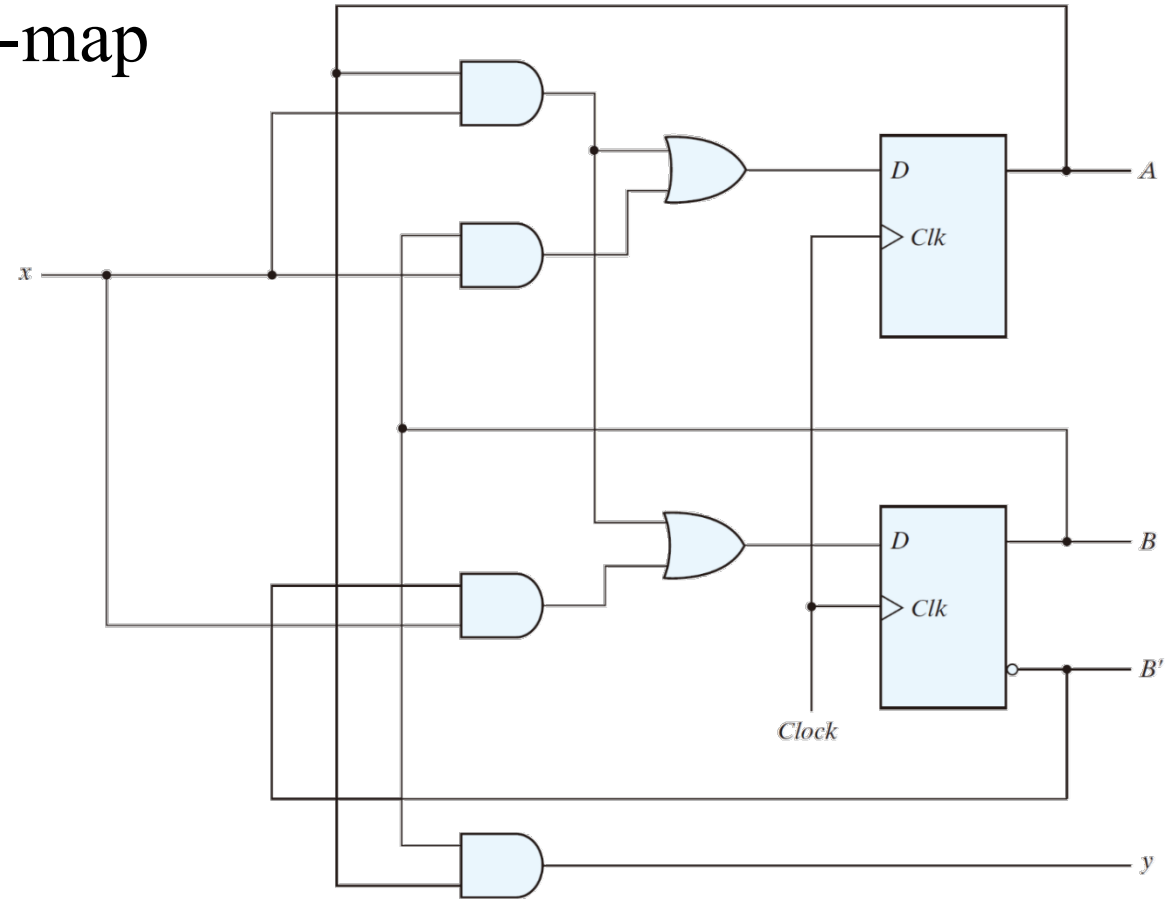


FIGURE 5.29

Logic diagram of a Moore-type sequence detector

5.8 Design Procedure (6/28)

Excitation Table of *JK* Flip-Flop

- A state diagram $\rightarrow$ State table $\rightarrow JK$ flip-flop input functions

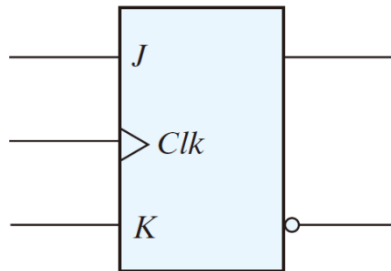


Table 5.1
Flip-Flop Characteristic Tables

<i>JK</i> Flip-Flop			
<i>J</i>	<i>K</i>	$Q(t + 1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	<i>J</i>	<i>K</i>
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

5.8 Design Procedure (7/28)

Excitation Table of T Flip-Flop

- State diagram $\rightarrow$ State table $\rightarrow T$ flip-flop input functions

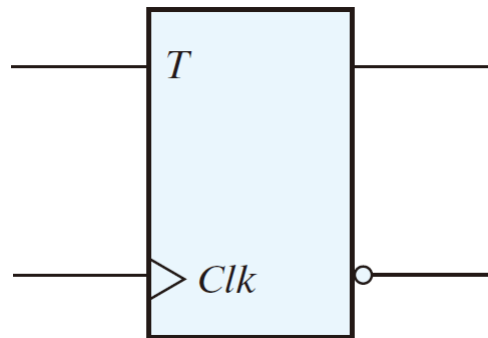


Table 5.1
Flip-Flop Characteristic Tables

T Flip-Flop		
T	$Q(t + 1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

5.8 Design Procedure (8/28)

Sequence Detector --- Synthesis Using D Flip-Flops

Problem → State diagram → State table → Equations → Circuit

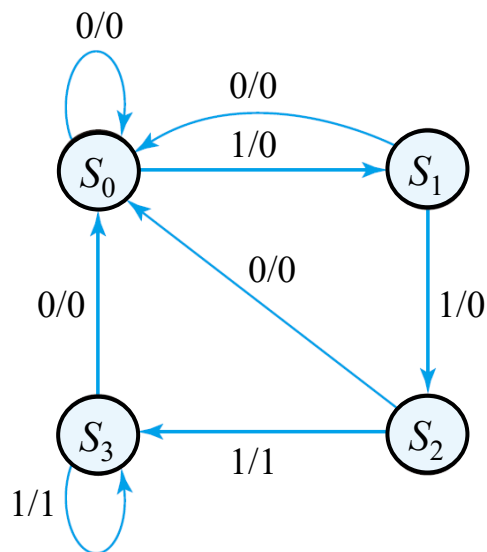


FIGURE 5.27

State diagram for sequence detector

Table 5.13

State Table and JK Flip-Flop Inputs

Present State		Input x	Next State		Flip-Flop Inputs				Output y
A	B		A	B	J_A	K_A	J_B	K_B	
0	0	0	0	0	0	X	0	X	0
0	0	1	0	1	0	X	1	X	0
0	1	0	1	0	1	X	X	1	0
0	1	1	0	1	0	X	X	0	0
1	0	0	1	0	X	0	0	X	0
1	0	1	1	1	X	0	1	X	0
1	1	0	1	1	X	0	X	0	1
1	1	1	0	0	X	1	X	1	1

Table 5.12

Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

5.8 Design Procedure (9/28)

Sequence Detector --- Synthesis Using *JK* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

Table 5.13
State Table and JK Flip-Flop Inputs

Present State		Input <i>x</i>	Next State		Flip-Flop Inputs				Output <i>y</i>
<i>A</i>	<i>B</i>		<i>A</i>	<i>B</i>	<i>J_A</i>	<i>K_A</i>	<i>J_B</i>	<i>K_B</i>	
0	0	0	0	0	0	X	0	X	0
0	0	1	0	1	0	X	1	X	0
0	1	0	1	0	1	X	X	1	0
0	1	1	0	1	0	X	X	0	0
1	0	0	1	0	X	0	0	X	0
1	0	1	1	1	X	0	1	X	0
1	1	0	1	1	X	0	X	0	1
1	1	1	0	0	X	1	X	1	1

• The *JK* flip-flop inputs

- $J_A(A, B, x) = \Sigma(2) + d(4, 5, 6, 7)$
- $K_A(A, B, x) = \Sigma(7) + d(0, 1, 2, 3)$
- $J_B(A, B, x) = \Sigma(1, 5) + d(2, 3, 6, 7)$
- $K_B(A, B, x) = \Sigma(2, 7) + d(0, 1, 4, 5)$

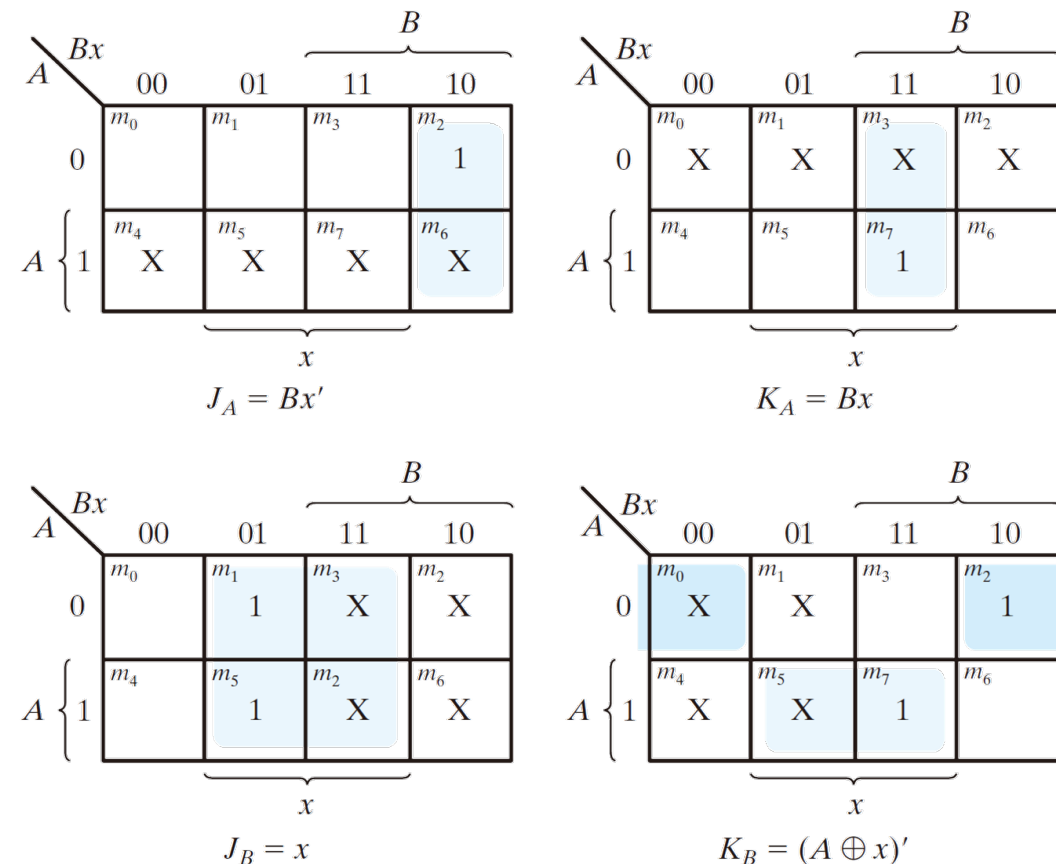


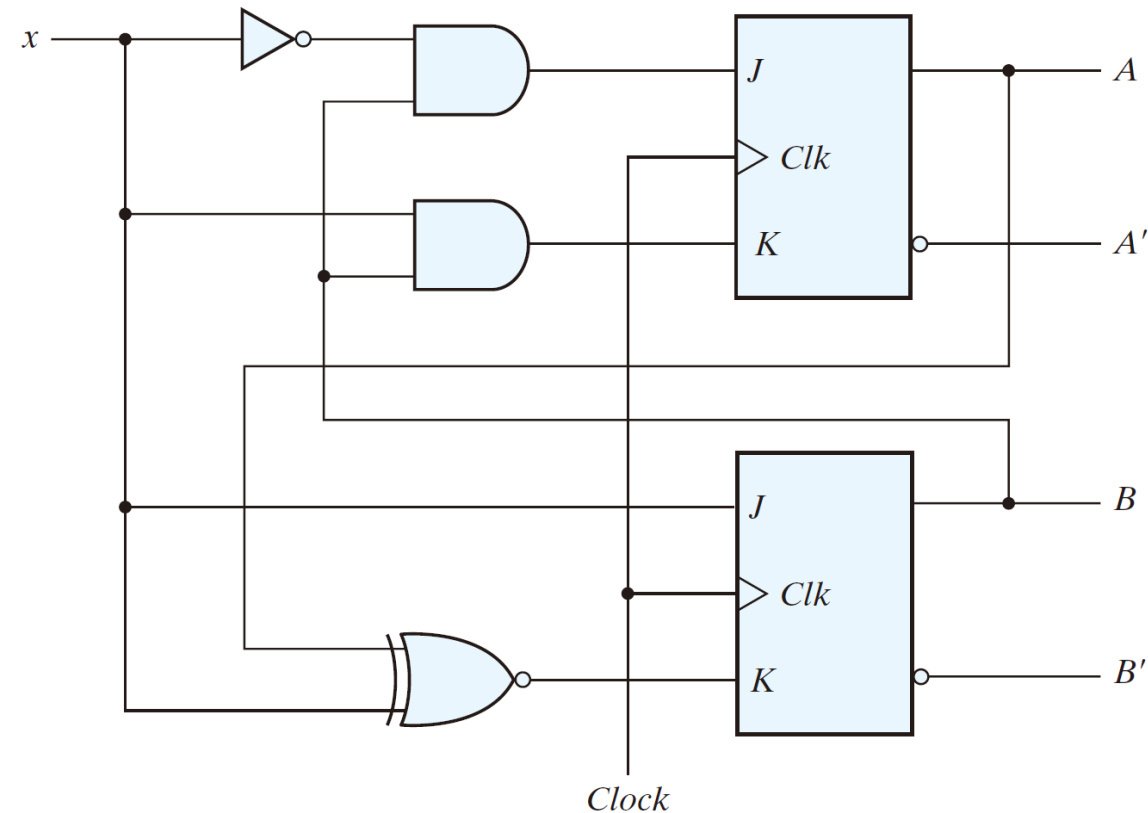
FIGURE 5.30
Maps for *J* and *K* input equations

5.8 Design Procedure (10/28)

Sequence Detector --- Synthesis Using *JK* Flip-Flops

Problem → State diagram → State table → Equations → Circuit

- $J_A = Bx'; K_A = Bx$
- $J_B = x; K_B = (A \oplus x)'$
- $y = AB$

**FIGURE 5.31**

Logic diagram for sequential circuit with J/K flip-flops

5.8 Design Procedure (11/28)

Synthesis Using T Flip-Flops

Problem → **State diagram** → State table → Equations → Circuit

- A 3-bit binary counter
 - No inputs (except for the clock input)

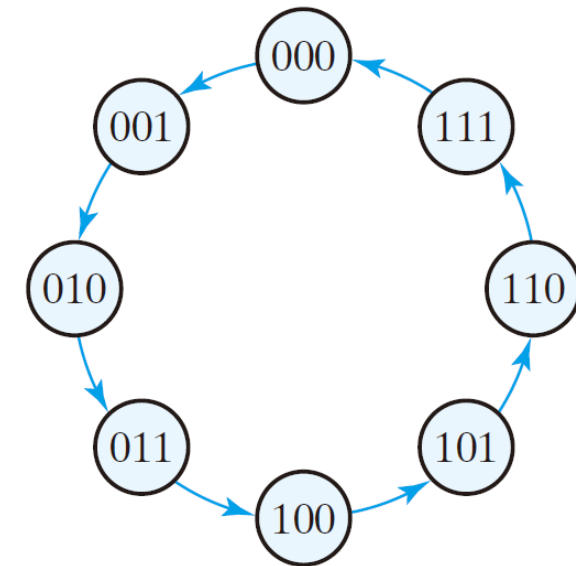


FIGURE 5.32

State diagram of three-bit binary counter

5.8 Design Procedure (12/28)

Synthesis Using T Flip-Flops

Problem → State diagram → State table → Equations → Circuit

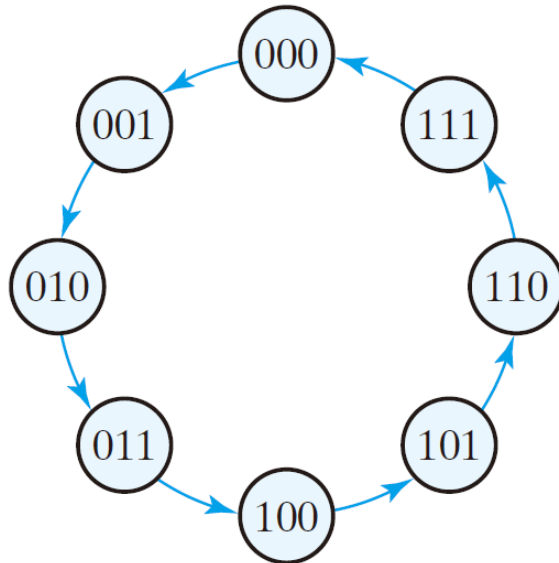


Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

Table 5.14
State Table for Three-Bit Counter

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	1	1
1	1	1	0	0	0	1	1	1

5.8 Design Procedure (13/28)

Synthesis Using T Flip-Flops

Problem → State diagram → State table → Equations → Circuit

Table 5.14
State Table for Three-Bit Counter

Present State			Next State			Flip-Flop Inputs		
A_2	A_1	A_0	A_2	A_1	A_0	T_{A2}	T_{A1}	T_{A0}
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	1	1
1	1	1	0	0	0	1	1	1

- The T flip-flop inputs
 - $T_{A2}(A_2, A_1, A_0) = \Sigma(3, 7)$
 - $T_{A1}(A_2, A_1, A_0) = \Sigma(1, 3, 5, 7)$
 - $T_{A0}(A_2, A_1, A_0) = \Sigma(0, 1, 2, 3, 4, 5, 6, 7)$

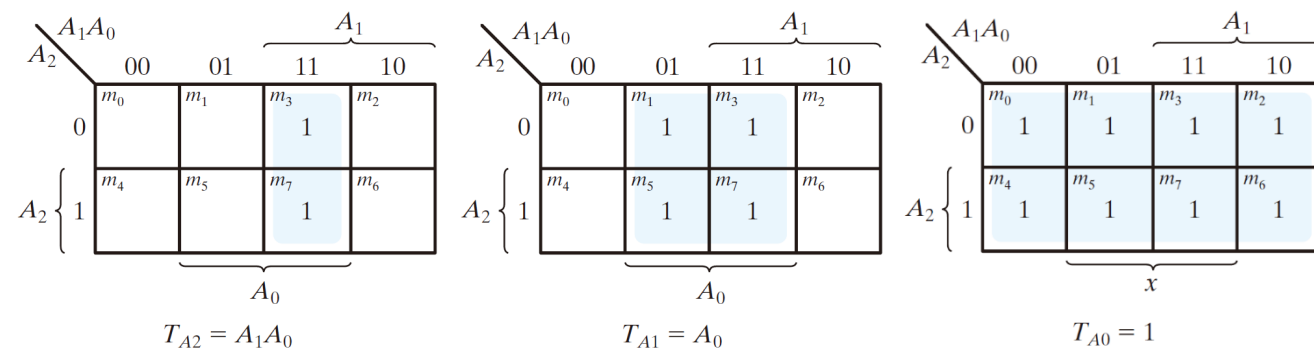


FIGURE 5.33
Maps for three-bit binary counter

5.8 Design Procedure (14/28)

Synthesis Using T Flip-Flops

Problem → State diagram → State table → Equations → Circuit

- Input equations

- $T_{A2} = A_1A_0$
- $T_{A1} = A_0$
- $T_{A0} = 1$

- The logic diagram

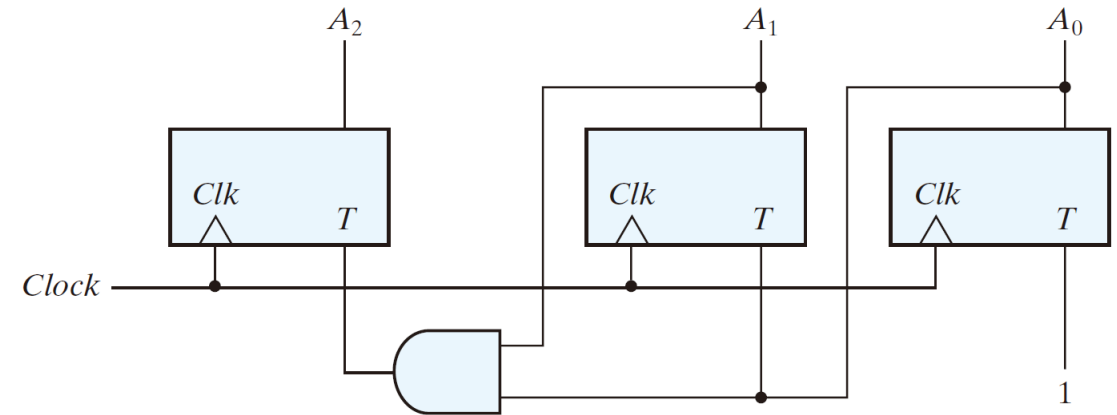


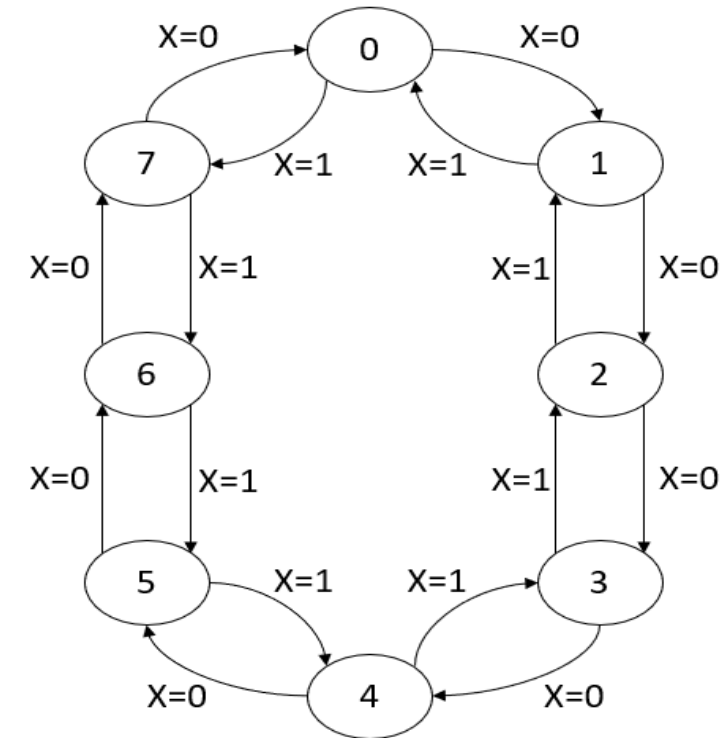
FIGURE 5.34
Logic diagram of three-bit binary counter

5.8 Design Procedure (15/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → **State diagram** → State table → Equation → Circuit

- 設計具有3個*JK*型正反器*A*, *B*, *C*與1個輸入*X*之3-bit counter。
 - $X=0$ 時，電路會上數
 - $X=1$ 時，電路會下數

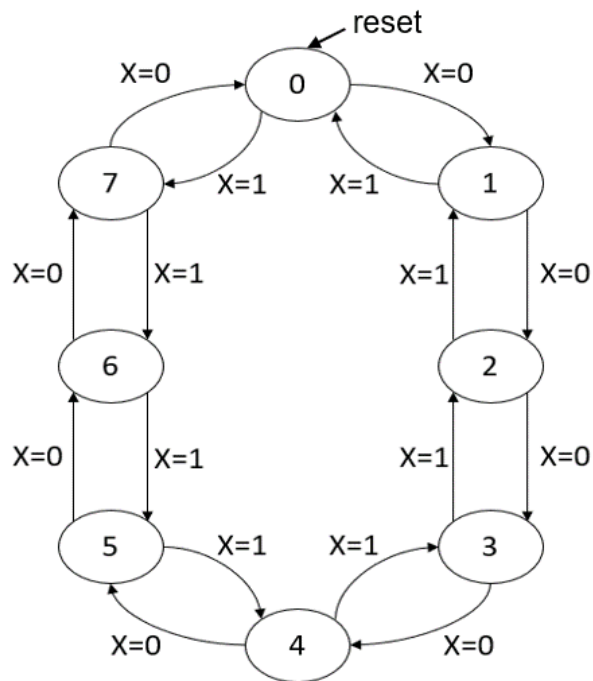


State diagram

5.8 Design Procedure (16/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit



State diagram

Present State				Next State		
<i>X</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>A</i>	<i>B</i>	<i>C</i>
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	1
0	0	1	1	1	0	0
0	1	0	0	1	0	1
0	1	0	1	1	1	0
0	1	1	0	1	1	1
0	1	1	1	0	0	0
1	0	0	0	1	1	1
1	0	0	1	0	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

State table

5.8 Design Procedure (17/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → **State table** → Equation → Circuit

Present State				Next State			Flip-Flop Inputs					
X	A	B	C	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	1	0	X	0	X	1	X
0	0	0	1	0	1	0	0	X	1	X	X	1
0	0	1	0	0	1	1	0	X	X	0	1	X
0	0	1	1	1	0	0	1	X	X	1	X	1
0	1	0	0	1	0	1	X	0	0	X	1	X
0	1	0	1	1	1	0	X	0	1	X	X	1
0	1	1	0	1	1	1	X	0	X	0	1	X
0	1	1	1	0	0	0	X	1	X	1	X	1
1	0	0	0	1	1	1	1	X	1	X	1	X
1	0	0	1	0	0	0	0	X	0	X	X	1
1	0	1	0	0	0	1	0	X	X	1	1	X
1	0	1	1	0	1	0	0	X	X	0	X	1
1	1	0	0	0	1	1	X	1	1	X	1	X
1	1	0	1	1	0	0	X	0	0	X	X	1
1	1	1	0	1	0	1	X	0	X	1	1	X
1	1	1	1	1	1	0	X	0	X	0	X	1

Table 5.12
Flip-Flop Excitation Tables

Q(t)	Q(t = 1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(a) *JK* Flip-Flop

5.8 Design Procedure (18/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

Present State				Flip-Flop Inputs					
X	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	X	0	X	1	X
0	0	0	1	0	X	1	X	X	1
0	0	1	0	0	X	X	0	1	X
0	0	1	1	1	X	X	1	X	1
0	1	0	0	X	0	0	X	1	X
0	1	0	1	X	0	1	X	X	1
0	1	1	0	X	0	X	0	1	X
0	1	1	1	X	1	X	1	X	1
1	0	0	0	1	X	1	X	1	X
1	0	0	1	0	X	0	X	X	1
1	0	1	0	0	X	X	1	1	X
1	0	1	1	0	X	X	0	X	1
1	1	0	0	X	1	1	X	1	X
1	1	0	1	X	0	0	X	X	1
1	1	1	0	X	0	X	1	1	X
1	1	1	1	X	0	X	0	X	1

State table

BC XA	00	01	11	10
00			1	
01	X	X	X	X
11	X	X	X	X
10	1			

$$J_A = XB'C' + X'BC$$

BC XA	00	01	11	10
00	X	X	X	X
01			1	
11	1			
10	X	X	X	X

$$K_A = XB'C' + X'BC$$

5.8 Design Procedure (19/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

Present State				Flip-Flop Inputs					
X	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	X	0	X	1	X
0	0	0	1	0	X	1	X	X	1
0	0	1	0	0	X	X	0	1	X
0	0	1	1	1	X	X	1	X	1
0	1	0	0	X	0	0	X	1	X
0	1	0	1	X	0	1	X	X	1
0	1	1	0	X	0	X	0	1	X
0	1	1	1	X	1	X	1	X	1
1	0	0	0	1	X	1	X	1	X
1	0	0	1	0	X	0	X	X	1
1	0	1	0	0	X	X	1	1	X
1	0	1	1	0	X	X	0	X	1
1	1	0	0	X	1	1	X	1	X
1	1	0	1	X	0	0	X	X	1
1	1	1	0	X	0	X	1	1	X
1	1	1	1	X	0	X	0	X	1

State table

BC XA	00	01	11	10
00		1	X	X
01		1	X	X
11	1		X	X
10	1		X	X

$$J_B = XC' + X'C$$

BC XA	00	01	11	10
00	X	X	1	
01	X	X	1	
11	X	X		1
10	X	X		1

$$K_B = XC' + X'C$$

5.8 Design Procedure (20/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

Present State				Flip-Flop Inputs					
X	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	0	X	0	X	1	X
0	0	0	1	0	X	1	X	X	1
0	0	1	0	0	X	X	0	1	X
0	0	1	1	1	X	X	1	X	1
0	1	0	0	X	0	0	X	1	X
0	1	0	1	X	0	1	X	X	1
0	1	1	0	X	0	X	0	1	X
0	1	1	1	X	1	X	1	X	1
1	0	0	0	1	X	1	X	1	X
1	0	0	1	0	X	0	X	X	1
1	0	1	0	0	X	X	1	1	X
1	0	1	1	0	X	X	0	X	1
1	1	0	0	X	1	1	X	1	X
1	1	0	1	X	0	0	X	X	1
1	1	1	0	X	0	X	1	1	X
1	1	1	1	X	0	X	0	X	1

State table

BC XA	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$$J_C = 1$$

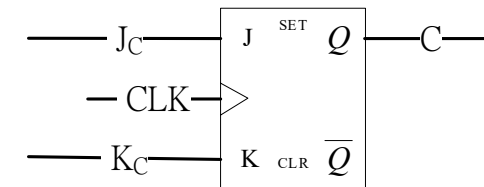
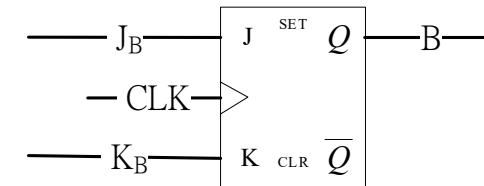
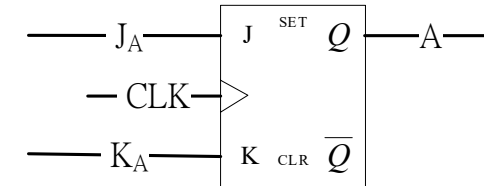
BC XA	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$$K_C = 1$$

5.8 Design Procedure (21/28)

Analysis with *JK* Flip-Flops: Example 3

Problem → State diagram → State table → Equation → Circuit

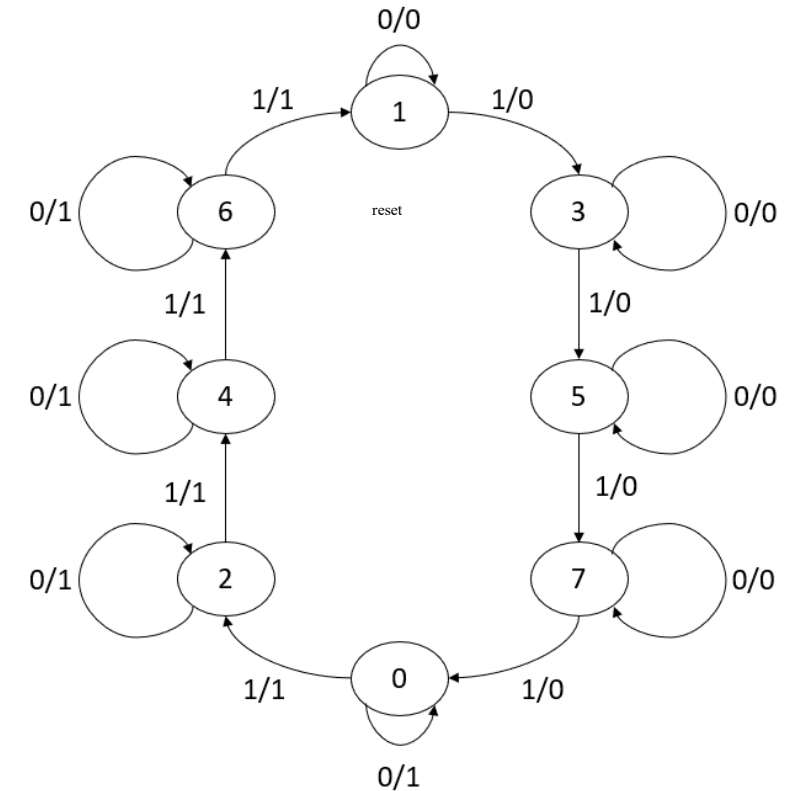


5.8 Design Procedure (22/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → **State diagram** → State table → Equation → Circuit

- 設計具有3個*JK*型正反器*A, B, C*、1個輸入*X*與1個輸出*Y*之3-bit counter。
 - $X=0$ 時，狀態值保持不變
 - $X=1$ 時，
 - 目前狀態值為奇數時，
 - 若目前狀態值為7，次一狀態值為0。
 - 否則，狀態值加2。
 - 目前狀態值為偶數時，
 - 若目前狀態值為6，次一狀態值為1。
 - 否則，狀態值加2。
 - 目前狀態值為奇數時， $Y=0$ ；為偶數時， $Y=1$

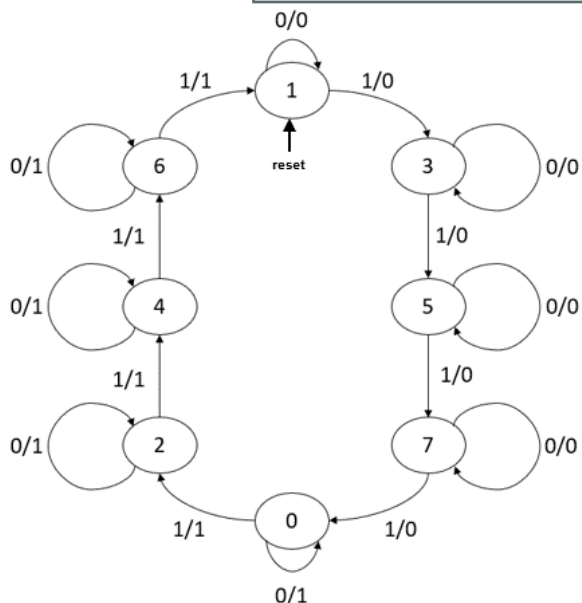


State diagram

5.8 Design Procedure (23/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit



State diagram

Table 5.12
Flip-Flop Excitation Tables

$Q(t)$	$Q(t = 1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(a) *JK* Flip-Flop

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

5.8 Design Procedure (24/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

BC XA	00	01	11	10
00				
01	X	X	X	X
11	X	X	X	X
10			1	1

$$J_A = XB$$

BC XA	00	01	11	10
00	X	X	X	X
01				
11			1	1
10	X	X	X	X

$$K_A = XB$$

5.8 Design Procedure (25/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

BC XA	00	01	11	10
00			X	X
01			X	X
11	1	1	X	X
10	1	1	X	X

$$J_B = X$$

BC XA	00	01	11	10
00	X	X	0	0
01	X	X	0	0
11	X	X	0	0
10	X	X	0	0

$$K_B = 0$$

5.8 Design Procedure (26/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

BC XA	00	01	11	10
00		X	X	
01		X	X	
11		X	X	1
10		X	X	

$$J_C = XAB$$

BC XA	00	01	11	10
00	X			X
01	X			X
11	X		1	X
10	X			X

$$K_C = XAB$$

5.8 Design Procedure (27/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → Circuit

Present State				Next State				Flip-Flop Inputs					
X	A	B	C	A	B	C	Y	JA	KA	JB	KB	JC	KC
0	0	0	0	0	0	0	1	0	X	0	X	0	X
0	0	0	1	0	0	1	0	0	X	0	X	X	0
0	0	1	0	0	1	0	1	0	X	X	0	0	X
0	0	1	1	0	1	1	0	0	X	X	0	X	0
0	1	0	0	1	0	0	1	X	0	0	X	0	X
0	1	0	1	1	0	1	0	X	0	0	X	X	0
0	1	1	0	1	1	0	1	X	0	X	0	0	X
0	1	1	1	1	1	1	0	X	0	X	0	X	0
1	0	0	0	0	1	0	1	0	X	1	X	0	X
1	0	0	1	0	1	1	0	0	X	1	X	X	0
1	0	1	0	1	0	0	1	1	X	X	0	0	X
1	0	1	1	1	0	1	0	1	X	X	0	X	0
1	1	0	0	1	1	0	1	X	0	1	X	0	X
1	1	0	1	1	1	1	0	X	0	1	X	X	0
1	1	1	0	0	0	1	1	X	1	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	0	X	1

State table

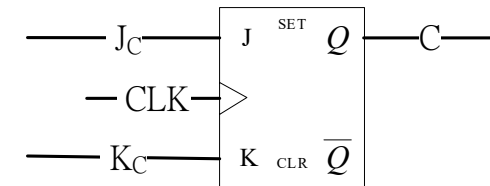
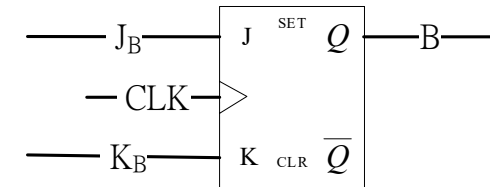
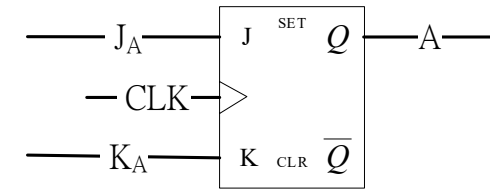
BC XA	00	01	11	10
00	1			1
01	1			1
11	1			1
10	1			1

$$Y = C'$$

5.8 Design Procedure (28/28)

Analysis with *JK* Flip-Flops: Example 4

Problem → State diagram → State table → Equation → **Circuit**



Y